PRACTICE SHEET

1. What is the interval in which the function $f(x)=\sqrt{9-x^2}$ is increasing? (f (x) > 0)
(a) $0 < x < 3$ (b) $-3 < x < 0$
(c) $0 < x < 9$ (d) $-3 < x < 3$

2. A wire 34 cm long is to be bent in the form of a quadrilateral of which each angle is 90°. What is the maximum area which can be enclosed inside the quadrilateral?
(a) $68cm^2$ (b) $70cm^2$
(c) $71.25\ cm^2$ (d) $72.25\ cm^2$

3. Which one of the following is correct? The function $f(x) = (x - 1)\ e^x + 1$ is
(a) Negative for all x > 0
(b) Positive for all x > 0
(c) Increasing for all x
(d) Decreasing for all x

4. The motion of a particle is described as $s = 2 - 3t + 4t^3$. What is the acceleration of the particle at the point where its velocity is zero?
(a) 0 (b) 4
(c) 8 unit (d) 12 unit

5. What is the product of two parts of 20, such that the product of one part and the cube of the other is maximum?
(a) 75 (b) 91
(c) 84 (d) 96

6. What is the maximum slope of the curve $y = -x^3 + 3x^2 + 2x - 27$?
(a) 1 (b) 2
(c) 5 (d) –23

7. What is the area of the largest rectangular field which can be enclosed with 200 m of fencing?
(a) $1600\ m^2$ (b) $2100\ m^2$
(c) $2400\ m^2$ (d) $2500\ m^2$

8. What is the smallest value of m for which $f(x) = x^2 + mx + 5$ is increasing in the interval $1 \le x \le 2$?
(a) m = 0 (b) m = – 1
(c) m = –2 (d) m = –3

9. What is the maximum value of x. y subject to the condition x + y = 8?
(a) 8 (b) 16
(c) 24 (d) 32

10. What is the equation of the curve whose slope at any point is equal to $2x$ and which passes through the origin?
(a) $y(1 - x) = x^2$ (b) $y^2(1+ x^2) = x^4$
(c) y2 = (x + 1) (d) y = x2

11. What is the maximum value of the function $\log x - x$?
(a) – 1 (b) 0
(c) 1 (d) ∞

12. A rectangular box with a cover is to have a square base. The volume is to be 10 cubic cm. The surface area of the box in terms of the side x is given by which one of the following function?
(a) $f(x) = (40 / x) + 2x^2$
(b) $f(x) = (40 / x) + x^2$
(c) $f(x) = (40 / x) + x^2$
(d) f (x) = (60 / x) + 2x

13. $f(x) = \cos x$ is monotonic decreasing under which one of the following conditions?
(a) $0 < x < \frac{\pi}{2}$ only (b) $\frac{\pi}{2} < x < \pi$ only
(c) $0 < x < \pi$ (d) $0 < x < 2\pi$

14. What is the minimum value of $2x^2 - 3x + 5$?
(a) 0 (b) 3/4
(c) 31/4 (d) 31/8

15. **Assertion (A):** The tangent to the curve $y = x^3 - x^2 - x + 2$ at (1, 1) is parallel to the x – axis.
Reason (R): The slope of the tangent to the curve at (1, 1) is zero
(a)Both A and R are true and R is the correct explanation of A
(b)Both A and R are true but R is not the correct explanation of A
(c)A is true but R is false
(d)A is false but R is false

16. The function $f(x) = x^2 - 2x$ increases for all
(a) $x > -1$ only (b) $x < -1$ only
(c) x > 1 only (d) x < 1 only

17. Let a and b be two distinct roots of a polynomial equation $f(x) = 0$. Then there exists at least one root lying between a and b of the polynomial equation.
(a) $f(x) = 0$ (b) $f'(x) = 0$
(c) $f''(x) = 0$ (d) None of these

18. The profit function, in rupees, of a firm selling x items ($x \ge 0$) per week is given by $P(x) = -3500 + (400 - x)x$. How many items should the firm sell so that the firm has maximum profit?
(a) 400 (b) 300
(c) 200 (d) 200

19. A stone thrown vertically upward satisfies the equation $s = 64t - 64t^2$, where s is in meter and t is in second. What is the time required to reach the maximum height?
(a) 1s (b) 2s
(c) 3s (d) 4s

20. If $f(x) = 3x^2 + 6x - 9$, then
(a) $f(x)$ is increasing in (–1, 3)
(b) $f(x)$ is decreasing in (3, ∞)
(c) $f(x)$ is increasing in (–∞, –1)
(d) f (x) is decreasing in (–∞, –1)

21. What is the value of p for which the function $f(x) = p \sin x + \frac{1}{3}\sin 3x$ has an extreme at $x = \frac{\pi}{3}$?
(a) 0 (b) 1

(c) – 1 (d) 2

22. If at any instant t, for a sphere, r denotes the radius, S denotes the surface area and V denotes the volume, then what is $\frac{dV}{dt}$ equal to?

(a) $\frac{1}{2}S\frac{dr}{dt}$ (b) $\frac{1}{2}r\frac{dS}{dt}$

(c) $r\frac{dS}{dt}$ (d) $\frac{1}{2}r^2\frac{dS}{dt}$

23. What is the interval over which the function: $f(x) = 6x - x^2$, $x > 0$ is increasing?

(a) (0, 3) (b) (3, 6)

(c) (6, 9) (d) (2, 3)

24. For a point of inflection of y = f(x), which one of the following is correct?

(a) $\frac{dy}{dx}$ must be zero

(b) $\frac{d^2y}{dx^2}$ must be zero

(c) $\frac{dy}{dx}$ must be non-zero

(d) $\frac{d^2y}{dx^2}$ must be non-zero

25. What is the least value of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on [–2, 2.5]?

(a) –3 (b) 8

(c) –19 (d) –16.5

26. If $f(x)= kx^3 - 9x^2 + 9x + 3$ is monotonically increasing in every interval, then which one of the following is correct?

(a) $k < 3$ (b) $k \le 3$

(c) $k > 3$ (d) $k \ge 3$

27. A balloon is pumped at the rate of $4cm^3/s$. What is the rate at which its surface area increases when its radius is 4 cm?

(a) $1cm^2/s$ (b) $2cm^2/s$

(c) $3cm^2/s$ (d) $4cm^2/s$

28. The growth of a quantity N(t) at any instant t is given by $\frac{dN(t)}{dt} = \alpha N(t)$.If $N(t) = ce^{kt}$, c is a constant, then what is the value of α?

(a) c (b) k

(c) c+k (d) c–k

29. What is the maximum point on the curve $x = e^x y$?

(a) (1,e) (b) $(1, e^{-1})$

(c) (e,1) (d) (e–1,1)

30. Given two squares of sides x and y such that $y = x + x^2$. What is the rate of change of area of the second square with respect to the area of the first square?

(a) $1+3x+2x^2$ (b) $1+2x+3x^2$

(c) $1-2x+3x^2$ (d) $1-2x-3x^2$

ANSWER KEYS

1.	b	**2.**	d	**3.**	c	**4.**	d	**5.**	a	**6.**	c	**7.**	d	**8.**	c	**9.**	b	**10.**	d
11.	a	**12.**	a	**13.**	c	**14.**	d	**15.**	d	**16.**	c	**17.**	b	**18.**	c	**19.**	b	**20.**	d
21.	d	**22.**	b	**23.**	a	**24.**	a	**25.**	c	**26.**	c	**27.**	b	**28.**	b	**29.**	b	**30.**	a

Solutions

Sol.1. (b)

f(x)= $\sqrt{9-x^2}$

f'(x)= $\frac{1}{2\sqrt{9-x^2}}\times(-2x)=-\frac{x}{\sqrt{9-x^2}}$

For function to be increasing

$-\frac{x}{\sqrt{9-x^2}}>0$

But $\sqrt{9-x^2}$ is defined only when

$9-x^2>0$ or $x^2-9<0$

$(x+3)(x-3)<0$

i.e. $-3<x<3<3$

$-3<x<3\cap x<0$

$\Rightarrow -3<x<0$

Sol.2. (d)

Let one side of quadrilateral be x and another side by y

So, 2 (x + y) = 34

or, (x + y) = 17

We know from the basic principle that for a given perimeter square has the maximum area, so, x = y and putting this value in equation (i)

x=y= $\frac{17}{2}$

Area=x.y= $\frac{17}{2}\times\frac{17}{2}=\frac{289}{4}$ =72.25

Sol.3. (c)

The given function is

$f(x)=(x-1)e^x+1$

$\Rightarrow f'(x)=(x-1)e^x+e^x$

$=(x-1+1)e^x=xe^x$

Thus, it is clear that f (x) is increasing for all x

Sol.4. (d)

Given rule is:

Distance, $s=2-3t+4t^3$

$\Rightarrow$Velocity= $\frac{ds}{dt}=-3+12t^2$

$\Rightarrow$Acceleration= $\frac{d^2s}{dt^2}=24t$

Since, velocity zero

$\therefore \frac{ds}{dt}=0$

$\Rightarrow 0=-3+12t^2\Rightarrow t=\sqrt{\frac{3}{12}}=\frac{1}{2}$

Acceleration (when velocity is zero)

$\Rightarrow \frac{d^2s}{dt^2}=24t=24\times\frac{1}{2}=12$ unit

Sol.5. (a)

Let 20 be divided in two parts such that

First part = x

$\therefore$ Second part = 20 – x

Now, assume that

$P=x^3(20-x)$

$=20x^3-x^4$

Now, $\frac{dp}{dx}=60x^2-4x^3$

and $\frac{d^2p}{dx^2}=120x-12x^2$

Put $\frac{dp}{dx}=0$ for maxima or minima

$\Rightarrow \frac{dp}{dx}=0$

$\Rightarrow 4x^2(15-x)=0\Rightarrow x=0,\ x=15$

$\therefore \left(\frac{d^2p}{dx^2}\right)_{x=15}$ =120×15–12×(225)

1800–2700= – 900<0

$\therefore$ P is a maximum at x = 15

$\therefore$ First part = 15

And second part = 20 – 15 = 5

Required product = 15 × 5 = 75

Sol.6. (c)

The equation of curve is given as:

$y=-x^3+3x^2+2x-27$

On differentiating w. r. t. x we get

$\frac{dy}{dx}=-3x^2+6x+2$

This represents slope of the curve at any point.

Let A= $\frac{dy}{dx}=-3x^2+6x+2$

$\Rightarrow \frac{dA}{dx}=-6x+6$ and $\frac{d^2A}{dx^2}=-6$

Put $\frac{dA}{dx}=0$ for maxima or minima

-6x + 6 = 0

$\Rightarrow$x=1

Now, $\left(\frac{d^2A}{dx^2}\right)_{x=1}$ 1= – 6 < 0

$\therefore$ A is maximum at x = 1

$\therefore$ Maximum slope of curve = – 3 + 6 + 2 = 5

Sol.7. (d)

Area of largest rectangular field for a given perimeter for is possible if length and breadth of rectangular field are equal i.e. it is a square

$\Rightarrow 4x=200\Rightarrow x=\frac{200}{4}=50m$

$\therefore$ Area of largest rectangular field = 50× 50 = 250 m^2.

Alter:

Let length and breadth of rectangular field be x and y respectively

$\therefore$ 2 (x + y) = 200

$\Rightarrow$ y = 100 – x

And area, A = xy

$=x(100-x)=100x-x^2$

$\because \frac{dA}{dx}=0$ for maxima or minima

Put $\frac{dA}{dx}$ =0 for maxima or minima

100 – 2x = 0

$\Rightarrow$ x = 50 $\Rightarrow$ y = 50

Now, which shows maximum, independent of values of x and y, b'. only when they are equal

$\therefore$A is maximum at x = 50.

Sol.8. (c)

$f(x)=x^2+mx+5$

Consider, option (c) m = –2

$f(x)=x^2-2x+5\Rightarrow f'(x)=2x-2$

It is clear that f' (x) > 0 in interval $1\le x\le 2$.

$\therefore$ m = – 2

Sol.9. (b)

The given condition is:

x + y = 8 $\Rightarrow$ y = 8 – x

and let, the product of x and y be,

P = xy

$\Rightarrow P=x(8-x)=8x-x^2$

Differentiating w.r.t.x

$\Rightarrow \frac{dp}{dx}=8-2x$ and $\frac{d^2P}{dx^2}=-2<0$

put $\frac{dp}{dx}$ =0, for maxima or minima

8–2x = 0

$\Rightarrow$x=8/2 = 4 and y = 4

And $\left(\frac{d^2p}{dx^2}\right)<0$ when x = y

$\therefore$ P is maximum at x = 4

Maximum value of P = 4. (8 – 4) = 4.4 = 16

Sol.10. (d)

Slope at the point $(x, y)=2x$

$\therefore \frac{dy}{dx}=2x$

$\Rightarrow \int dy=\int 2x\,dx$

$\Rightarrow y=x^2+c$

Given that, it passes through the origin.

So, $0=(0)^2+c$

$\Rightarrow$ c = 0 then, $y=x^2$

Sol.11. (a)

Let $y=\log x-x$

$\therefore \frac{dy}{dx}=\frac{1}{x}-1=0\Rightarrow\frac{1}{x}=1\Rightarrow x=1$

For maximum and minimum value of y

$\frac{dy}{dx}=\frac{1}{x}-1=0\Rightarrow\frac{1}{x}=1\Rightarrow x=1$

For x=1, $\frac{d^2y}{dx^2}$ = –ve

Thus, the value of given function is maximum for $x=1$

So, the maximum value of the function = log (1) – 1 = – 1

Sol.12. (a)

Let the height of rectangular box be y cm.

$\therefore$ Volume = $x\times x\times y$

$\Rightarrow y=\frac{10}{x^2}$..........(i)

Now, surface area of box

$=2(x^2+xy+yx)=2(x^2+2xy)$

$=2\left(x^2+\frac{20}{x}\right)$ From equation (i)

$=2x^2=\frac{40}{x}$

Sol.13. (c)

$\Rightarrow f'(x)=-\cos x$,

Now $f'(x) = 0$

$\Rightarrow -\sin x = 0$

$\Rightarrow \sin x = \sin(0)$

$\Rightarrow x = n\pi$

Clearly, $f'(x) < 0$, when $0 < x < \pi$

Hence f (x) is decreasing when $0 < x < \pi$

Sol.14. (d)

Let $y = 2x^2 - 3x + 5$

$\Rightarrow \frac{dy}{dx} = 4x - 3$ and $\frac{d^2y}{dx^2} = 4$

For maximum and minimum value of y

$\frac{dy}{dx} = 0$

$\Rightarrow 4x - 3 = 0$

$\Rightarrow x = 3/4$

since, for every value of x, $\frac{d^2y}{dx^2}$ = +ve

so, minimum value of $2x^2 - 3x + 5$ is minimum x=3/4

$\therefore$ the minimum value $= 2\left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{4}\right) + 5$

$= \frac{9}{8} - \frac{9}{4} + 5 = \frac{9 - 18 + 40}{8} = \frac{31}{8}$

Sol.15. (d)

Given equation of the curve is $y = x^3 - x^2 - x + 2$

$\frac{dy}{dx} = 3x^2 - 2x - 1$

$\Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = 3 - 2 - 1 = 0$

Since, $\frac{dy}{dx}$ at (1,1) is 0

$\therefore$ Slope of the tangent=0

Hence,the equation of tangent at (1,1) is

$y - 1 = 0(x - 1) \Rightarrow y = 1$

ie, parallel to x – axis.

Both A and R true and R is the correct explanation of A.

Sol.16. (c)

Given $f(x) = x^2 - 2x$

On differentiating w.r.t.'x'we get $f'(x) = 2x - 2$

f (x) is increasing, if $f'(x) > 0$

$\Rightarrow 2x - 2 > 0$

$\Rightarrow x > 1$

Sol.17. (b)

Rolle's theorem says between any two roots of a polynomial *f* (x), there is always a root of its derivative $f'(x)$

Therefore between a and b. There exist at least one root of the polynomial equation $f'(x) = 0$

Sol.18. (c)

$p(x) = -3500 + (400 - x)x = -3500 + 400x - x^2$

On differentiating w.r.t.x. we get

$P'(x) = 400 - 2x$

Put $P'(x) = 0$ for maxima or minima

$\Rightarrow 400 - 2x = 0$

$\Rightarrow x = 200$

Now $P''(x) = -2x$

$\Rightarrow P''(200) = -400 < 0$

$\therefore$ P (x) is maximum at x = 200

Hence 200 items should the firm sell so that the firm has maximum profit.

Sol.19. (b)

Given equation is $s = 64t - 16t^2$

$\therefore$ On differentiating w. r. t. *t*, we get

$\frac{ds}{dt} = 64 - 32t$

Put $\frac{ds}{dt} = 0$ for maximum height

$\Rightarrow 64 - 32t = 0 \Rightarrow t = 2$

Now, $\frac{d^2s}{dt^2} = -32$

At t=2, $\frac{d^2s}{dt^2} = -32$

Since, $\left(\frac{d^2s}{dt^2}\right)_{t=2} < 0$

$\therefore$ Required time=2 second

Sol.20. (d)

Given $f(x) = 3x^2 + 6x - 9$

On differentiating w. r. t. x, we get

$f'(x) = 6x + 6$

$f'(x) < 0$

$\Rightarrow 6x + 6 < 0$

$\Rightarrow 6x < -6$

$\Rightarrow x < -1$

Hence f(x) is decreasing in $(-\infty, -1)$

Sol.21. (d)

Given function is, $f(x) = p\sin x + 1/3\sin 3x$

$f'(x) = p\cos x + \cos 3x$

But given, that, f(x) is maximum at $x = \pi/3$

From equation (i),

$p\cos\pi/3 + \cos\pi = 0$

$\Rightarrow p/2 - 1 = 0 \Rightarrow p = 2$

Sol.22. (b)

$\because S = 4\pi r^2$

$\Rightarrow \frac{dS}{dt} = 8\pi r.\frac{dr}{dt}$, and $V = \frac{4}{3}\pi r^2$

$\Rightarrow \frac{dV}{dt} = \frac{4}{3}\pi.3r^2\frac{dr}{dt} = 4\pi r^2\frac{dr}{dt} = \frac{4\pi r^2}{8\pi r}.\frac{dS}{dt} = \frac{1}{2}r.\frac{dS}{dt}$

Sol.23. (a)

$f'(x) = 6 - 2x$

f(x) > will be increasing function, if

$f'(x) > 0 \Rightarrow 6 - 2x > 0 \Rightarrow x < 0$

Thus, required interval is (0,3)

Sol.24. (a)

Sol.25. (c)

$y = 2x^3 - 3x^2 - 12x + 1$

$y' = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 0$

$x = 2, -1$

$y'' = 12x - 6$ at x = 2 it is positive

minima at x = 2 ; y(2) = –19

y(-2) = –3 minimum value –19

Sol.26. (c)

$\because f(x) = kx^3 - 9x^2 + 9x + 3$

On differentiating w.r.t.x., we get

$f'(x) = 3kx^2 - 18x + 9$

For a function to be monotonically increasing

$\Delta = b^2 - 4ac < 0$

$\Rightarrow 36 - 12k < 0 \Rightarrow k > 3$

Sol.27. (b)

Let r be the radius of balloon.

$\therefore V = \frac{4}{3}\pi r^2 \qquad \Rightarrow \frac{dV}{dt} = \frac{4}{3}\pi.3r^2\frac{dr}{dt}$

$\Rightarrow 4 = \frac{4}{3}\pi.3(4)^2\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{16\pi}$

Now, $S = 4\pi r^2$

$\Rightarrow \frac{dS}{dt} = 4\pi.2r.\frac{dt}{dt} = 4\pi \times 2 \times 4 \times \frac{1}{16\pi} = 2cm^2/s$

Sol.28. (b)

$\because N(t) = ce^{kt}$

$\therefore \frac{dN(t)}{dt} = \frac{d}{dt}ce^{kt} = k(ce^{kt}) = k[N(t)]$

But $\frac{dN(t)}{dt} = \alpha N(t) \qquad \therefore \alpha = k$

Sol.29. (b)

$\because x = e^x$ y and $y = xe^{-x}$ $\therefore \frac{dy}{dx} = e^{-1}(1 - x)$

Now, $\frac{dy}{dx} = 0 \qquad \Rightarrow e^{-1}(1 - x) = 0$

$\Rightarrow x = 1 \qquad \because \frac{d^2y}{dx^2} < 0$

$\therefore$ Maximum value at x = 1 and y = 1. $e^{-1} = e^{-1}$

Sol.30. (a)

$\because$ Area of first square, $A_1 = x^2$, and area of second

Square, $A_2 = y^2 = (x + x^2) = x^2 + x^4 + 2x^3$

Now, $\frac{dA_1}{dx} = 2x$, and $\frac{dA_2}{dx} = 2x + 4x^3 + 6x^2$

Hence, $\frac{dA_2}{dA_1} = \frac{2x + 4x^3 + 6x^2}{2x} = 1 + 2x^2 + 3x$

NDA PYQ

1. The function $f(x) = k \sin x + 1/3 \sin 3x$ has maximum value at $x = \pi/3$, what is the value of k?
(a) 3 (b) 1/3
(c) 2 (d) 1/2
[NDA (I) - 2011]

2. The largest value of $2x^3 - 3x^2 - 12x + 5$ for $-2 \le x \le 2$ occurs when:
(a) $x = -2$ (b) $x = -1$
(c) $x = 2$ (d) $x = 0$
[NDA (I) - 2011]

3. Consider the following statements in respect of the function $f(x) = x^3 - 1$, $x \in [-1,1]$
1. f(x) is increasing in [-1,1]
2. f(x) has no root in (-1,1)
Which of the above statements is/are correct ?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (I) - 2011]

4. The function y=f(x)=mx+c has:
(a)maximum point but no minimum point
(b)minimum point but no maximum point
(c)both maximum and minimum points
(d)neither maximum point nor minimum point
[NDA-2011(2)]

5. At an extreme point of a function f(x), the tangent to the curve is:
(a) parallel to the x-axis
(b) Perpendicular to the x-axis
(c) Inclined at an angle 45° to the x-axis
(d) Inclined at an angle 60° to the x-axis
[NDA-2011(2)]

6. The point in the interval $(0, 2\pi)$ where $f(x) = e^x \sin x$ has maximum slope is:
(a) $\pi/4$ (b) $\pi/2$
(c) π (d) $3\pi/2$
[NDA-2011(2)]

7. If the rate of change in volume of spherical soap bubble is uniform, then the rate of change of surface area varies as:
(a) square of radius
(b) square root of radius
(c) inversely proportional to radius
(d) cube of the radius
[NDA-2011(2)]

8. If $f(x) = x \ln x$, then f(x) attains minimum value at which one of the following points?
(a) $x=e^{-2}$ (b) x=c
(c) $x=e^{-1}$ (d) $x=2e^{-1}$
[NDA-2011(2)]

9. What is the slope of the tangent to the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at t=2 ?
(a) 7/6 (b) 6/7
(c) 1 (d) 5/6
[NDA (I) - 2012]

10. Which on the following statement is correct?
(a)The derivative of a function f(x) at a point will exist if there is one tangent to the curve y=f(x)at the point and the tangent is parallel to y-axis
(b)The derivative of a function f(x) at a point will exist if there is one tangent to the curve y=f(x) at the point and the tangent must be parallel to y-axis
(c)The derivative of a function f(x) at a point will exist if there is one and only one tangent to the curve y = f(x) at the point and the tangent is not parallel to y-axis
(d)None of these
[NDA (I) - 2012]

11. How many tangents are parallel to x axis for the curve $y = x^2 - 4x + 3$?
(a) 1 (b) 2
(c) 3 (d) 0
[NDA (I) - 2012]

12. The function $f(x) = x^3 - 3x^2 + 6$ is an increasing function for
(a) $0 < x < 2$ (b) $x < 2$
(c) $x > 2$ or $x < 0$ (d) all x
[NDA (II) - 2012]

13. What is the minimum value of $|x|$?
(a) –1 (b) 0
(c) 1 (d) 2
[NDA (II) - 2012]

14. The radius of a circle is uniformly increasing at the rate of 3cm/s. What is the rate of increase in area, when the radius is 10cm?
(a) 6π cm²/s (b) 10π cm²/s
(c) 30π cm²/s (d) 60π cm²/s
[NDA-2012(2)]

15. The curve $y = xe^x$ has minimum value equal to:
(a) $-1/e$ (b) $1/e$
(c) $-e$ (d) e
[NDA (I) - 2013]

16. Consider the following statements:
I.The derivative where the function attains maxima or minima be zero
II.If a function is differentiable at a point, then it must be continuous at the point.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2013]

17. The function $f(x) = x^2 - 4x$, $x \in [0,4]$ attains minimum value at:
(a) x = 0 (b) x = 1
(c) x = 2 (d) x = 4
[NDA (I) - 2013]

18. The minimum value of the function $f(x) = |x-4|$ exists at:
(a) x = 0 (b) x = 2
(c) x = 4 (d) x = – 4
[NDA (II) - 2013]

19. The maximum value of the function $f(x) = x^3 + 2x^2 - 4x + 6$ exists at:
(a) x= – 2 (b) x = 1
(c) x = 2 (d) x = –1
[NDA (II) - 2013]

(for next two): Consider the function

$$f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$$

20. What is the minimum value of the function?

(a) 1/2 (b) 1/3
(c) 2 (d) 3

[NDA (I) - 2014]

21. What is the maximum value of the function?
(a) 1/2 (b) 1/3
(c) 2 (d) 3

[NDA (I) - 2014]

Directions (for next two): Consider the curve $y = e^{2x}$

22. What is the slope of the tangent to the curve at (0,1)?
(a) 0 (b) 1
(c) 2 (d) 4

[NDA (I) - 2014]

23. Where does the tangent to the curve at (0,1) meet the X-axis?
(a) (1,0) (b) (2,0)
(c) $\left(-\frac{1}{2}, 0\right)$ (d) $\left(\frac{1}{2}, 0\right)$

[NDA (I) - 2014]

Directions (for next two): Consider an ellipse
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

24. What is the area of the greatest rectangle that can be inscribed in the ellipse ?
(a) ab (b) 2ab
(c) ab/2 (d) $\sqrt{ab}$

[NDA (I) - 2014]

25. What is the area included between the ellipse and the greatest rectangle inscribed in the ellipse?
(a) ab(π - 1) (b) 2ab(π - 1)
(c) ab(π - 2) (d) none of these

[NDA (I) - 2014]

26. What is the slope of the tangent to the curve $y=\sin^{-1}(\sin^2 x)$ at x=0?
(a) 0 (b) 1
(c) 2 (d) None

[NDA-2014(1)]

Directions (for next two): Read the following information carefully and answer the question given below.
A cylinder is inscribed in a sphere of radius r.

27. What is the height of the cylinder of maximum volume?
(a) $\frac{2r}{\sqrt{3}}$ (b) $\frac{r}{\sqrt{3}}$
(c) r (d) $\sqrt{3}r$

[NDA (II) - 2014]

28. What is the radius of the cylinder of maximum volume?
(a) $\frac{2r}{\sqrt{3}}$ (b) $\frac{\sqrt{2}r}{\sqrt{3}}$
(c) r (d) $\sqrt{3}r$

[NDA (II) - 2014]

Directions (for next two): Read the following information carefully and answer the questions given below:
A rectangular box is to be made from a sheet of 24 inch length and 9 inch width cutting out identical squares of side length x from the four corners and turning up the sides.

29. What is the value of x for which the volume is maximum?
(a) 1 inch (b) 1.5 inch
(c) 2 inch (d) 2.5 inch

[NDA (II) - 2014]

30. What is the maximum volume of the box?
(a) 200cu inch (b) 400 cu inch
(c) 100cu inch (d) None of these

[NDA (II) - 2014]

Directions (for next two): Consider the function
$f(x) = 0.75x^4 - x^3 - 9x^2 + 7$

31. What is the maximum value of the function?
(a) 1 (b) 3
(c) 7 (d) 9

[NDA (I) - 2015]

32. Consider the following statements:
I.The function attain local minima at x = – 2 and x = 3.
II.The function increases in the interval (–2,0)
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2015]

Directions (for next two): Consider the function f(x)= $\frac{x^2-1}{x^2+1}$ where $x \in R$.

33. At what value of x does f(x) attain minimum value?
(a) –1 (b) 0
(c) 1 (d) 2

[NDA (I) - 2015]

34. What is the minimum value of f(x)?
(a) 0 (b) 1/2
(c) –1 (d) 2

[NDA (I) - 2015]

35. Consider the following statements
1. $y = \frac{e^x + e^{-x}}{2}$ is an increasing function on [0, ∞)
2. $y = \frac{e^x - e^{-x}}{2}$ is an increasing function on (–∞, ∞).
Which of the above statements is/are correct ?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2015]

Directions (33 to 34): Consider the function f(x) = $\left(\frac{1}{x}\right)^{2x^2}$ where x > 0.

36. At what value of x does the function attain the maximum value?
(a) e (b) $\sqrt{e}$
(c) $\frac{1}{\sqrt{e}}$ (b) $\frac{1}{e}$

[NDA (II) - 2015]

37. The maximum value of the function is:
(a) e (b) $e^{\frac{2}{e}}$
(c) $e^{\frac{1}{e}}$ (d) $\frac{1}{e}$

[NDA (II) - 2015]

Direction (for next two): Consider the function $f(x) = -2x^3 - 9x^2 - 12x + 1$

38. The function f(x) is a decreasing function in the interval.
(a) (–2, –1) (b) Only (–∞, –2)
(c) Only (–1, ∞) (d) (–∞, –2) ∪ (–1, ∞)

[NDA (II) - 2015]

39. The function f (x) is an increasing function is the interval

(a) $(-2,-1)$ (b) $(-\infty, -2)$
(c) $(-1, 2)$ (d) $(-1, \infty)$

[NDA (II) - 2015]

40. Consider the following statements

1.The function $f(x) = x^2 + 2\cos x$ is increasing in the interval (θ, π)

2.The function $f(x) = \ln\left(\sqrt{1+x^2} - x\right)$ is decreasing in the interval $(-\infty, \infty)$

Which of the above statements is/are correct ?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2015]

41. Consider the following statements?

1. $f(x) = \ln x$ is an increasing function on $(0,\infty)$.

2. $f(x) = e^x - x(\ln x)$ is an increasing function on $(1, \infty)$

Which of the above statements is/are correct ?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2015]

42. The function $f(x) = \frac{x^2}{e^x}$ is monotonically increasing if

(a) Only $x < 0$ (b) Only $x > 2$
(c) $0 < x < 2$ (d) $x \in (-\infty, 0) \cup (2, \infty)$

[NDA (II) - 2015]

For the next two (2) items that follow:

$f(x) = (x-1)^2 (x+1) (x-2)^3$

43. What is the number of points of local minima of the function f(x)?

(a) None (b) One
(c) Two (d) Three

[NDA (I) - 2016]

44. What is the number of points of local maxima of the function f(x)?

(a) None (b) One
(c) Two (d) Three

[NDA (I) - 2016]

For the next two items that follow:

consider the function

$f(x) = |x - 1| + x^2$ where $x \in R$

45. Which one of the following statements is correct?

(a)f(x) is increasing in $\left(-\infty, \frac{1}{2}\right)$ and decreasing in $\left(\frac{1}{2}, \infty\right)$

(b)f(x) is decreasing in $\left(-\infty, \frac{1}{2}\right)$ and increasing in $\left(\frac{1}{2}, \infty\right)$

(c)f(x) is increasing in $(-\infty, 1)$ and decreasing in $(1, \infty)$

(d)f(x) is decreasing in $(-\infty, 1)$ and increasing in $(1, \infty)$

[NDA (I) - 2016]

46. Which one of the following statements is correct?

(a)f(x) has local minima at more than one point in $(-\infty, \infty)$

(b)f(x)has local maxima at more than one point in $(-\infty, \infty)$

(c)f(x) has local minimum at one point only in $(-\infty, \infty)$

(d)f(x) has neither maxima nor minima in $(-\infty, \infty)$

[NDA (I) - 2016]

Directions (for next three): Consider the function

$f(\theta) = 4(\sin^2 \theta + \cos^4 \theta)$

47. What is the maximum value of the function $f(\theta)$?

(a) 1 (b) 2
(c) 3 (d) 4

[NDA (I) - 2016]

48. What is the minimum value of the function $f(\theta)$?

(a) 0 (b) 1
(c) 2 (d) 3

[NDA (I) - 2016]

49. Consider the following statements:

1.$f(\theta)=2$ has no solution

2.$f(\theta)=7/2$ has a solution

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2016(1)]

For the next two (02) items that follow:

Consider the function

$f(x) = \frac{27\left(x^{2/3} - x\right)}{4}$

50. How many solution does the function f(x)=1 have?

(a) one (b) two
(c) three (d) four

[NDA-2016(1)]

51. How many solutions does the function f(x)= –1 have?

(a) one (b) two
(c) three (d) four

[NDA-2016(1)]

For the next (02) items that follow:

Consider the equation

$k \sin x + \cos 2x = 2k - 7$

52. If the equation possesses solution, then what is the minimum value of k?

(a) 1 (b) 2
(c) 4 (d) 6

[NDA-2016(1)]

53. If the equation possesses solution, then what is the maximum value of k?

(a) 1 (b) 2
(c) 4 (d) 6

[NDA-2016(1)]

54. Which one of the following statement is correct in respect of the function $f(x) = x^3 \sin x$?

(a) It has local maximum at x = 0

(b) It has local minimum at x = 0

(c) It has neither maximum or minimum at x = 0

(d) It has maximum value 1

[NDA (II) - 2016]

Consider the following for the next two items that follow:

$$f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \le x \le 2 \\ 37 - x, & 2 < x \le 3 \end{cases}$$

55. Which of the following is/are correct?

1. f(x) is increasing in the interval [-1,2]
2. f(x) is decreasing in the interval (2,3]

Select the correct answer using the code given below

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2016]

56. Which of the following statements are correct?

1. f(x) is continuous at x = 2
2. f(x) attains greatest value at x = 2
3. f(x) is differentiable at x = 2

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (II) - 2016]

Consider the following for the next two items that follow:

$f(x) = \begin{cases} \frac{e^x - 1}{x}, & x > 0 \\ 0, & x = 0 \end{cases}$ be a real valued function

57. Which of the following is/are correct?
(a)f(x) is strictly decreasing in the interval $(0,\infty)$
(b)f(x) is strictly increasing in the interval (0,x)
(c)f(x) is neither increasing nor decreasing in (0,x)
(d)f(x) is not decreasing in (0,x)

[NDA (II) - 2016]

58. Which of the following statements are correct?
1. f(x) is right continuous at x = 0
2. f(x) is discontinuous at x = 1
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA (II) - 2016]

59. What is the maximum area of a triangle that can be inscribed in a circle of radius a?
(a) $\frac{3a^2}{4}$ (b) $\frac{a^2}{2}$
(c) $\frac{3\sqrt{3}a^2}{4}$ (d) $\frac{\sqrt{3}a^2}{4}$

[NDA (I) - 2017]

60. What is the length of the longest interval in which the function $f(x) = 3\sin x - 4\sin^3 x$ is increasing?
(a) $\pi/3$ (b) $\pi/2$
(c) $3\pi/2$ (d) π

[NDA (I) - 2017]

61. What is the maximum value of the function $f(x) = 4\sin^2 x + 1$?
(a) 5 (b) 3
(c) 2 (d) 1

[NDA (I) - 2017]

62. Let f(x) = x + 1/x, where $x\in(0,1)$. Then which one of the following is correct?
(a) f(x) fluctuates in the interval
(b) f(x) increases in the interval
(c) f(x) decreases in the interval
(d) none of the above

[NDA-2017(1)]

63. Which one of the following is correct is respect of the function: f(x) = x (x–1) (x+1)?
(a)The local maximum value is larger than local minimum value
(b)The local maximum value is smaller than local minimum value
(c)The function has no local maximum
(d)The function has no local minimum

[NDA (II) - 2017]

64. Match list-I with list-II and select the correct answer using the codes given below the lists:

List-I (Function)	List-II Maximum Value)
A. Sinx + cosx	(i) $\sqrt{10}$
B. 3sin + 4cos x	(ii) $\sqrt{2}$
C. 2sinx + cos x	(iii)5
D. sin x + 3 cos x	(iv) $\sqrt{5}$

(a) A-ii, B-iii, C-i, D-iv
(b) A-ii, B-iii, C-iv, D-i
(c) A-iii, B-ii, C-i, D-iv
(d) A-iii, B-ii, C-iv, D-i

[NDA (II) - 2017]

65. The maximum value of $\frac{\ln x}{x}$ is:
(a) e (b) 1/e
(c) 1/e (d) 1

[NDA (II) - 2017]

66. A cylindrical jar without a lid has to be constructed using a given surface area of a metal sheet. If the capacity of the jar is to be maximum, then the diameter of the jar must be k times the height of the jar. The value of k is:
(a) 1 (b) 2
(c) 3 (d) 4

[NDA (II) - 2017]

67. Consider the following statements:
Statement I:
x > sin x for all x > 0
Statement II:
f(x) = x–sinx is an increase function for all x > 0
Which one of the following is correct respect of the above statement?
(a)both statement I and statement II are true and statement II is correct explanation of statement I
(b)both statement I and statement II are true and statement II is not correct explanation of statement I.
(c)statement I is true but statement II is false
(d)statement I is false but statement II is true

[NDA-2017(2)]

68. What is the maximum value of $16\sin\theta - 12\sin^2\theta$?

[NDA (I) - 2018]

(a) 3/4 (b) 4/3
(c) 16/3 (d) 4

Consider the following information for the next three (03) items that follow:

Three sides of a trapezium are each equal to 6cm. Let $\alpha\in\left(0, \frac{\pi}{2}\right)$ be the angle between a pair of adjacent sides.

69. If the area of the trapezium is the maximum possible, then what is α equal to?
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{2\pi}{5}$

[NDA (I) - 2018]

70. If the area of the trapezium is maximum, what is the length of the fourth side?
(a) 8cm (b) 9cm
(c) 10cm (d) 12cm

[NDA (I) - 2018]

71. What is the maximum area of the trapezium?
(a) $36\sqrt{3}$ cm^2 (b) $30\sqrt{3}$ cm^2
(c) $27\sqrt{3}$ cm^2 (d) $24\sqrt{3}$ cm^2

[NDA (I) - 2018]

72. What is the minimum value of $[x(x-1)+1]^{1/3}$, where $0 \le x \le 1$?
(a) $\left(\frac{3}{4}\right)^{\frac{1}{3}}$ (b) 1

(c) $\frac{1}{2}$ (d) $\left(\frac{3}{8}\right)^{\frac{1}{3}}$

[NDA (II) - 2018]

73. A flower in the form of a sector has been fenced by a wire of 40m length. If the flower-bed has the greatest possible area, then what is the radius of the sector?
(a) 25m (b) 20m
(c) 10m (d) 5m

[NDA (II) - 2018]

74. In which one of the following intervals is the function f(x) = $x^2 - 5x + 6$ decreasing.
(a) $(-\infty, 2)$ (b) $[3, \infty]$
(c) $(-\infty, \infty)$ (d) (2,3)

[NDA (II) - 2018]

75. Which one of the following is correct in respect of the function.

$f(x) = x \sin x + \cos x + \frac{1}{2}\cos^2 x$?

(a) It is increasing in the interval $\left(0, \frac{\pi}{2}\right)$
(b) It remain constant in the interval $\left(0, \frac{\pi}{2}\right)$
(c) It is decreasing in the interval $\left(0, \frac{\pi}{2}\right)$
(d) It is decreasing in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

[NDA (II) - 2018]

76. Let the slope of the curve $y = \cos^{-1}(\sin x)$ be $\tan\theta$. Then the value of θ in the interval $(0, \pi)$ is:
(a) $\frac{\pi}{6}$ (b) $\frac{3\pi}{4}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

[NDA (II) - 2018]

77. What is the minimum value of a^2x+b^2y, if $xy=c^2$
(a) abc (b) 2abc
(c) 3abc (d) 4abc

[NDA (I) - 2019]

78. The minimum distance from the point (4,2) to $y^2 = 8x$ is equal to
(a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) 2 (d) $3\sqrt{2}$

[NDA (I) - 2019]

79. A given quantity of metal is to be cast into a half cylinder (i.e. with a rectangular base and semicircular ends). if the total surface area is be minimum , then the ratio of the height of the half cylinder to the diameter of the semi circular ends is
(a) $\pi : (\pi + 2)$ (b) $\pi + 2): \pi$
(c) 1 : 1 (d)none of these

[NDA (I) - 2019]

80. What is the value of k for which the sum of the squares of the roots of $2x^2-2(k-2)x-(k+1)=0$ is minimum?
(a) –1 (b) 1
(c) 3/2 (d) 2

[NDA-2019(2)]

Direction (for next three):
A curve $y = me^{mx}$ where m > 0 intersects y-axis at a point P.

81. What is the slope of the curve at the point of intersection P?
(a) m (b) m^2
(c) 2m (d) $2m^2$

[NDA (II) - 2019]

82. How much angle does the tangent at P make with y - axis?
(a) $\tan^{-1} m^2$ (b) $\cot^{-1}(1+m^2)$
(c) $\sin^{-1}\left(\frac{1}{\sqrt{1+m^4}}\right)$ (d) $\sec^{-1}\sqrt{1+m^4}$

[NDA (II) - 2019]

83. What is the equation of tangent to the curve at P ?
(a) y = mx + m (b) y = −mx + 2m
(c) $y = m^2x + 2m$ (d) $y = m^2x + m$

[NDA (II) - 2019]

84. If $f(x) = \frac{x^3}{3} - \frac{5x^2}{2} + 6x + 7$ increases in the interval T and decreases in the interval S, then which one of the following is correct?
(a) $T = (-\infty,2)\cup(3,\infty)$ and S = (2,3)
(b) $T = \phi$ and $S = (-\infty,\infty)$
(c) $T = (-\infty,\infty)$ and $S = \phi$
(d) T = (2,3) and $S = (-\infty,2)\cup(3,\infty)$

[NDA (II) - 2019]

Direction (for next two)
Consider the function
$f(x) = 3x^4 - 20x^3 - 12x^2 + 288x + 1$

85. In which one of the following intervals in the function increasing?
(a) (−2,3) (b) (3,4)
(c) (−3, −2) (d) (−4, −3)

[NDA (II) - 2019]

86. In which one of the following intervals in the function decreasing?
(a) (−2,3) (b) (3,4)
(c) (4, 6) (d) (6, 9)

[NDA (II) - 2019]

87. What is the maximum value of sinx. cosx?
(a) 2
(b) 1
(c) 1/2
(d) limit does not exist

[NDA 2020]

88. What is the minimum value of $3\cos\left(A+\frac{\pi}{3}\right)$ where A∈R ?
(a) −3 (b) −1
(c) 0 (d) 3

[NDA 2020]

89. Consider the following statements:
1.The function flx) = In x increases in the interval $(0,\infty)$.
2.The function flx) = tan x increases in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
Which of the above statements is/are correct ?
(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

90. Let ℓ be the length and b be the breadth of a rectangle such that ℓ + b = k. What is the maximum area of the rectangle?
(a) $2k^2$ (b) k^2
(c) $k^2/2$ (d) $k^2/4$

[NDA 2020]

91. Let $y = 3x^2 + 2$. If x changes from 10 to 10.1, then what is the total change in y ?
(a) 4.71 (b) 5.23
(c) 6.03 (d) 8.01

[NDA 2020]

92. What is the minimum value of $|x-1|$, where $x \in R$?
(a) 0 (b) 1
(c) 2 (d) -1

[NDA 2020]

93. The radius of the circle is increasing at the rate of 0.7 cm/sec. What is the rate of increase of its circumference?
(a) 4.4 cm/sec (b) 8.4 cm/sec
(c) 8.8 cm/sec (d) 15.4 cm/sec

[NDA 2020]

94. Consider the following statements in respect of the function f(x) = sinx
1. f(x) increases in the interval $(0,\pi)$
2. f(x) decreases in the interval $\left(\frac{5\pi}{2}, 3\pi\right)$
Which of the above statements is/are correct ?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2021]

95. A 24 cm long wire is bent in the form a triangle with one of the angles as 60°. What is the altitude of the triangle having the greatest possible area?
(a) $4\sqrt{3}$ cm (b) $2\sqrt{3}$ cm
(c) 6 cm (d) 3 cm

[NDA (I) - 2021]

96. The curve $y = -x^3 + 3x^2 + 2x - 27$ has the maximum slope at
(a) x = -1 (b) x = 0
(c) x = 1 (d) x = 2

[NDA (I) - 2021]

97. If x + y = 20 and P = xy, then what is the maximum value of P ?
(a) 100 (b) 96
(c) 84 (d) 50

[NDA (I) - 2021]

98. What is the maximum value of sin2x.cos2x?
(a) 1/2 (b) 1
(c) 2 (d) 4

[NDA (I) - 2021]

99. What is the condition that $f(x) = x^3 + x^2 + kx$ has no local extremes?
(a) 4k < 1 (b) 3k > 1
(c) 3k < 1 (d) 3k ≤ 1

[NDA (II) - 2021]

100. If the function $f(x) = x^2 - kx$ is monotonically increasing in the interval $(1, \infty)$, then which one of the following is correct?
(a) k<2 (b) 2<k<3
(c) 3<k<4 (d) k>4

[NDA (II) - 2021]

101. The tangent to the curve $x^2 = y$ at (1, 1) makes an angle θ with the positive direction of x-axis. Which one of the following is correct?
(a) $\theta < \frac{\pi}{6}$ (b) $\frac{\pi}{6} < \theta < \frac{\pi}{4}$
(c) $\frac{\pi}{4} < \theta < \frac{\pi}{3}$ (d) $\frac{\pi}{3} < \theta < \frac{\pi}{2}$

[NDA (II) - 2021]

102. What is the slope of the tangent of $y = \cos^{-1}(\cos x)$ at $x = -\frac{\pi}{4}$?
(a) –1 (b) 0
(c) 1 (d) 2

[NDA (II) - 2021]

103. Consider the following statements in respect of the function $f(x) = x^2 + 1$ in the interval (1, 2):
1.The maximum value of the function is 5
2.The minimum value of the function is 2.
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2021]

104. Where does the tangent to the curve $y = e^x$ at the point (0, 1) meet x-axis?
(a) (1, 0) (b) (–1, 0)
(c) (2, 0) (d) $(-\frac{1}{2}, 0)$

[NDA (II) - 2021]

105. For what value of θ, where $0 < \theta < \frac{\pi}{2}$, does sin θ + sin θ cos θ attain maximum value?
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

[NDA (II) - 2021]

106. Consider the following statements in respect of the function $f(x) = x + \frac{1}{x}$:
1.The local maximum value of f(x) is less than its local minimum value
2.the local maximum value of f(x) occurs at x = 1
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA (II) 2021]

107. What is the maximum area of a rectangle what can be inscribed in a circle of radius 2 units?
(a) 4 square units
(b) 6 square units
(c) 8 square units
(d) 16 square units

[NDA (II) 2021]

108. Let p, q and 3 be respectively the first, third and fifth terms of an AP. Let d be the common difference. If the product (pq) is minimum, then what is the value of d?
(a) 1 (b) $\frac{3}{8}$
(c) $\frac{9}{8}$ (d) $\frac{9}{4}$

[NDA (II) 2021]

109. How many extreme values does sin4x + 2x, where $0 < x < \frac{\pi}{2}$ have?
(a) 1 (b) 2
(c) 4 (d) 8

[NDA (I) - 2022]

110. What is the maximum value of the function f(x)= $\frac{1}{\tan x + \cot x}$, where $0 < x < \frac{\pi}{2}$?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) 1 (d) 2
[NDA (I) - 2022]

111. In which one of the following intervals in the function f(x) $\frac{x^3}{3} - \frac{7x^2}{2} + 6x + 5$ decreasing?
(a) $(-\infty, 1)$ only (b) (1, 6)
(c) $(6, \infty)$ only (d) $(-\infty, 1) \cup (6, \infty)$
[NDA (I) - 2022]

112. If xy = 4225 where x, y are natural numbers, then is the minimum value of x + y ?
(a) 130 (b) 260
(c) 2113 (d) 4226
[NDA (I) - 2022]

113. Under which one of the following conditions does the function $f(x) = (p \sec x)^2 + (q \operatorname{cosec} x)^2$ attain minimum value?
(a) $\tan 2x = \frac{q}{p}$ (b) $\cot 2x = \frac{q}{p}$
(c) tan2x= pq (d) cot2 x = pq
[NDA 2022 (II)]

114. Where does the function f(x) = $\sum_{j=1}^{7}(x-j)^2$ attain its minimum value?
(a) x=3.5 (b) x = 4
(c) x = 4.5 (d) x = 5
[NDA 2022 (II)]

115. Consider the following statements in respect of the function
$$f(x) = \begin{cases} |x| + 1 & 0 < |x| \le 3 \\ 1, & x = 0 \end{cases}$$
1. The function attain maximum value only at x = 3
2. The function attain local minimum only at x = 0

Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA 2022 (II)]

116. What is the minimum value of the function $f(x) = \log_{10}(x^2 + 2x + 11)$?
(a) 0 (b) 1
(c) 2 (d) 10
[NDA 2022 (II)]

117. Consider the following statement:
1.f(x) = *l*nx is increasing in $(0, \infty)$
2.$g(x) = e^x + e^{\frac{1}{x}}$ is decreasing in $(0, \infty)$
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) Neither 1 nor 2
[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:
Given that $m(\theta) = \cot^2\theta + n^2\tan^2\theta + 2n$, where n is a fixed positive real number.

118. What is the least value of $m(\theta)$?
(a) n (b) 2n
(c) 3n (d) 4n
[NDA – 2023 (1)]

119. Under what condition does m attain the least value?
(a) $n = \tan^2\theta$ (b) $n = \cot^2\theta$
(c) $n = \sin^2\theta$ (d) $n = \cos^2\theta$
[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:
Consider the function f(x) = |x–2|+ |3–x| + |4–x|, where $x \in R$.

120. At what value of x does the function attain minimum value ?
(a) 2 (b) 3
(c) 4 (d) 0
[NDA – 2023 (1)]

121. What is the minimum value of the function?
(a) 2 (b) 3
(c) 4 (d) 0
[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:
Let x = $\frac{\sin^2 A + \sin A + 1}{\sin A}$ where $0 < A \le \frac{\pi}{2}$

122. What is the minimum value of x?
(a) 1 (b) 2
(c) 3 (d) 4
[NDA – 2023 (1)]

123. At what value of A does x attain the minimum values?
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
[NDA – 2023 (1)]

Consider the following for the next (02) items that follow
A function is defined by
$$f(x)= \begin{vmatrix} x+1 & 2 & 3 \\ 2 & x+4 & 6 \\ 3 & 6 & x+9 \end{vmatrix}$$

124. The function is decreasing on:
(a) $\left[-\frac{28}{3}, 0\right]$ (b) $\left[0, \frac{28}{3}\right]$
(c) $\left[0, \frac{50}{3},\right]$ (d) $\left[0, \frac{56}{3}\right]$
[NDA-2023 (2)]

125. The function attains local minimum value at:
(a) x= –28/3 (b) x= –1
(c) x=0 (d) x=28/3
[NDA-2023 (2)]

Consider the following for the next (02) items that follow
Given that $4x^2 + y^2 = 9$

126. What is the maximum value of y?
(a) 3/2 (b) 3
(c) 4 (d) 6
[NDA-2023 (2)]

127. What is the maximum value of xy?
(a) 9/4 (b) 3/2
(c) 4/9 (d) 2/3
[NDA-2023 (2)]

128. What is the greatest value of the function f(x) = $\sqrt{2-x} + \sqrt{2+x}$?

(a) √3 (b) √6
(c) √8 (d) 4

[NDA-2023 (2)]

Consider the following for the next two (02) items that follow:

A differentiable function f(x) has a local maximum at x = 0. Let y = 2f(x) + ax – b.

129. Which of the following is/are correct?

1. f′ (0) = 0
2. f″(0) < 0

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA-2024 (1)]

130. The function y has a relative maxima at x = 0 for

(a) a > 0, b = 0
(b) for all b and a = 0
(c) for all b > 0 only
(d) for all a and b = 0

[NDA-2024 (1)]

131. The non-negative values of b for which the function $\frac{16x^3}{3} - 4bx^2 + x$ has neither maximum nor minimum in the range x > 0 is

(a) 0 < b < 1 (b) 1 < b < 2
(c) b > 2 (d) 0 ≤ b < 1

[NDA-2024 (1)]

Consider the following for the next two (02) items that follow:

Let $f(x) = \frac{x}{\ln x}; (x > 1)$

132. Consider the following statements:

1. f(x) is increasing in the interval (e, ∞)
2. f(x) is decreasing in the interval (1, 3)
3. 9ln7 > 7ln9

Wchih of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2024 (1)]

133. Consider the following statements

1. $f''(e) = \frac{1}{e}$
2. f(x) attains local minimum value at x = e
3. A local minimum value of f(x) is e

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

Let f(x) = cos2x + x on [–π/2 , π/2]

134. What is the greatest value of f(x) ?

(a) $\frac{\sqrt{3}}{2} - \frac{\pi}{12}$ (b) $\frac{\sqrt{3}}{2} + \frac{\pi}{12}$

(c) $\frac{\sqrt{3}}{2} + \frac{\pi}{9}$ (d) $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$

[NDA-2024 (2)]

135. What is the least value of f(x)?

(a) $-\left(1 + \frac{\pi}{2}\right)$ (b) $-\left(\frac{1}{2} + \frac{\pi}{2}\right)$

(c) $-\left(1 + \frac{\pi}{4}\right)$ (d) $-2\left(\frac{1}{2} - \frac{\pi}{4}\right)$

[NDA-2024 (2)]

136. What is the minimum value of the function f(x) = $\log_{10}(x^2 + 2x + 11)$?

(a) 0 (b) 1
(c) 2 (d) 10

[NDA-2024 (2)]

Consider the following for the two (02) items that follow:

Let ABC be a right-angled at B and AB + AC = 3 units.

137. What is $\angle A$ equal to if the area of the triangle is maximum?

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{5\pi}{12}$

[NDA-2025 (1)]

138. What is the maximum area of the triangle?

(a) √3 /2 sq unit (b) √3 sq unit
(c) √6 /2 sq unit (d) √6 sq unit

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the function f(x) = x^2 + 9

139. Consider the following statements:

I. f(x) is an increasing function.
II. f(x) has local maximum at x = 0.

Which of the statements given above is/are correct?

(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2025 (1)]

For the following **one** items:

Consider the function f(x) = x|x|

140. Consider the following statements:

I. The function is increasing in the interval (– ∞, ∞).
II. The function is differentiable at x = 0.

Which of the statements given above is/are correct?

(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

For the following next **one** items:

Consider the function $f(x) = 1 - \sqrt[3]{(x-1)^2}$

141. The function has

(a) a minimum at x = 1
(b) a maximum at x = 1
(c) neither maximum nor minimum at x = 1
(d) no extremum

[NDA-2025 (2)]

142. A wire of length 20 cm is to be bent into a rectangle.

Which of the following statements is/are correct?

I. The rectangle of the largest area is the square.
II. It is possible to form a rectangle of an area of 27 cm².

Which of the statements given above is/are correct?

(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

143. Consider the following statements regarding the function $f(x) = \frac{1}{x-5}$

Statement–I:

f(x) is decreasing on the intervals (x < 5) and (x > 5).

Statement–II:

f '(x) > 0 for all $x \neq 5$

Which one of the following is correct in respect of the above statements?

(a) Both Statement–I and Statement–II are correct and Statement–II explains Statement–I

(b) Both Statement–I and Statement–II are correct but Statement–II does not explain Statement–I

(c) Statement–I is correct but Statement–II is not correct

(d) Statement–I is not correct but Statement–II is correct

[NDA-2025 (2)]

144. Consider the following statements:

Statement–I:

The function $f(x)=\frac{x^3+128}{x}$ has a minimum value 48 at x = 4.

Statement–II:

As x increases through 4, f '(x) changes sign from positive to negative.

Which one of the following is correct in respect of the above statements?

(a) Both Statement–I and Statement–II are correct and Statement–II explains Statement–I

(b) Both Statement–I and Statement–II are correct but Statement–II does not explain Statement–I

(c) Statement–I is correct but Statement–II is not correct

(d) Statement–I is not correct but Statement–II is correct

[NDA-2025 (2)]

ANSWER KEY

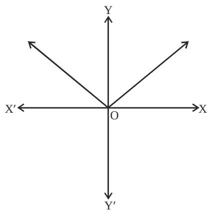
1.	c	**2.**	b	**3.**	c	**4.**	d	**5.**	a	**6.**	b	**7.**	c	**8.**	c	**9.**	b	**10.**	c
11.	a	**12.**	c	**13.**	b	**14.**	d	**15.**	a	**16.**	c	**17.**	c	**18.**	c	**19.**	a	**20.**	b
21.	d	**22**	c	**23.**	c	**24.**	b	**25.**	c	**26.**	a	**27.**	a	**28.**	b	**29.**	c	**30.**	a
31.	c	**32.**	c	**33.**	b	**34.**	c	**35.**	c	**36.**	c	**37.**	c	**38.**	d	**39.**	a	**40.**	c
41.	c	**42.**	c	**43.**	c	**44.**	b	**45.**	b	**46.**	c	**47.**	d	**48.**	d	**49.**	c	**50.**	b
51.	a	**52.**	b	**53.**	d	**54.**	b	**55.**	c	**56.**	a	**57.**	b	**58.**	d	**59.**	c	**60.**	a
61.	a	**62.**	c	**63.**	a	**64.**	b	**65.**	b	**66.**	b	**67.**	b	**68.**	c	**69.**	c	**70.**	d
71.	c	**72.**	a	**73.**	c	**74.**	a	**75.**	a	**76.**	b	**77.**	b	**78.**	c	**79.**	a	**80.**	c
81.	b	**82.**	c	**83.**	d	**84.**	a	**85.**	a	**86.**	b	**87.**	c	**88.**	a	**89.**	c	**90.**	d
91.	c	**92.**	a	**93.**	a	**94.**	b	**95.**	a	**96.**	c	**97.**	a	**98.**	a	**99.**	b	**100.**	a
101.	d	**102.**	a	**103.**	d	**104.**	b	**105.**	b	**106.**	a	**107.**	c	**108.**	c	**109.**	b	**110.**	b
111.	b	**112.**	a	**113.**	a	**114.**	b	**115.**	b	**116.**	b	**117.**	a	**118.**	d	**119.**	b	**120.**	b
121.	a	**122.**	c	**123.**	d	**124.**	a	**125.**	c	**126.**	b	**127.**	a	**128.**	c	**129.**	c	**130.**	b
131.	d	**132.**	d	**133.**	b	**134.**	b	**135.**	a	**136.**	b	**137.**	c	**138.**	a	**139.**	d	**140.**	c
141.	b	**142.**	a	**143.**	c	**144.**	c												

Solutions

Sol.1. (c)
Given function is, f(x) = k sin x +1/3 sin 3x
f′(x) = k cos x + cos 3x
But given, that, f(x) is maximum at x = π/3 from equation (i),
k cos π/3 + cos π = 0
⇒k/2– 1 = 0 ⇒ k = 2
Sol.2. (b)
Let $f(x) = 2x^3 – 3x^2 – 12x + 5$
$f'(x) = 6x^2 – 6x – 12$
For largest value, f′(x) = 0
$\Rightarrow 6x^2 – 6x – 12 = 0 \quad \Rightarrow x^2 – x – 2 = 0$
⇒x = – 1, 2
f″(x) = 12x – 6 ⇒ x = – 1,2
At x = 2, f″(2) = 24 – 6 = 18 > 0 (minimum)
At x = – 1, f″(–1) = – 12 – 6 = – 18 < 0 (maximum)
So, the function is maximum (largest) at x = – 1, and its largest value is:
$f(–1) = 2 (–1)^3 – 3 (–1)^2 – 12(–1) + 5$
= – 2 – 3 + 12 + 5 = 12
Sol.3. (c)
$y = x^3 - 1$
$y'=3x^2$ it is always positive so function is always increasing
$y = x^3 - 1 = 0$
x = 1 so no root exist in (-1.1)
Sol.4. (d)
y = mx + c
dy/dx = m
slope of tangent is not zero, there will be neither maxima nor minima
Sol.5. (a)
at extreme point tangent is parallel to x axis whose slope is always zero.
Sol.6. (c)
$f(x) = e^x \sin x$
$f'(x) = \text{slope} = e^x(\sin x – \cos x)$
for maximum slope differentiation of slope is zero
$f''(x) = e^x(\sin x – \cos x) + e^x(\cos x + \sin x)$
$f''(x) = 2e^x \sin x = 0$
⇒sinx = 0
⇒x = 0 or π
Sol.7. (d)
Rate of change in volume is uniform i.e. change in volume is directly proportion to unit power of r
$\frac{dV}{dt} \propto r$
$\Rightarrow \frac{dA}{dt} \propto r^2$
$\Rightarrow \frac{dr}{dt} \propto r^3$
Sol.8. (c)
Y = x logx
For minimum value dy/dx = 0
dy/dx = 1 + logx = 0
logx = - 1
$x = e^{-1}$
Sol.9. (b)
$x = t^2 +3t-8$, $y = 2t^2 -2t -5$
dx/dt = 2t + 3,at t = 2 dx/dt = 7
dy/dt = 4t-2 , at t = 2 dy/dt =6
dy/dx = 6/7

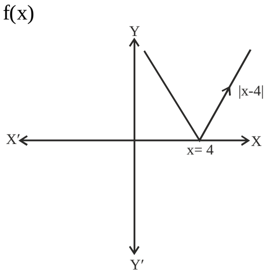
Sol.10. (c)
The derivative of a function f(x) at a point will exist if there is one and only one tangent to the curve y = f(x) at the point and the tangent is not parallel to y-axis
Sol.11. (a)
$y = x^2 - 4x + 3$
dy/dx = 2x - 4 = 0
x = 2 only one tangent
Sol.12. (c)
$f(x) = x^3 - 3x^2 + 6$
for increasing $f'(x) = 3x^2 - 6x > 0$
3x(x - 2) > 0
x > 2 or x < 0
Sol.13. (b)
Let y = |x|

$$y = \begin{bmatrix} x, & x \geq 0 \\ -x, & x < 0 \end{bmatrix}$$



So, the minimum value of |x| is zero
Sol.14. (a)
$A = \pi r^2$
$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(3)(10) = 60\pi$
Sol.15. (a)
Given curve, $y = xe^x$
On differentiating w.r.t.x., we get
$\frac{dy}{dx} = x . e^x + e^x . 1 = xe^x + e^x$
From maximum and minimum of y,
$\frac{dy}{dx} = 0 \Rightarrow e^x(x+1) = 0$
$\Rightarrow x = – 1 \ (\because e^x \neq 0)$
Again, differentiating w.r.t.x., we get
$\frac{d^2y}{dx^2} = x. e^x + e^x.1 = xe^x + 2e^x$
$\left(\frac{d^2y}{dx^2}\right)_{at\,x=-1} = (–1)e^{–1} + 2e^{–1} = \frac{1}{e} > 0$
(minimum)
∴ f(x) have minimum value at x = – 1.
Hence, minimum value is $y(–1) = (–1)e^{–1} = \frac{-1}{e}$
Sol.16. (c)
I.The derivative where the function attains maxima or minima must be zero.
II. If a function is differentiable at a point, then it must be continuous at that point but if a function is continuous at a point, then it is not necessarily that function is differentiable at that point.
So, both statements are correct
Sol.17. (c)
Given function,
$f(x) = x^2 – 4x$, x∈[0,4]
On differentiating w.r.t.x., we get
f′(x) = 2x – 4

For max or min of f(x),
f′(x) = 0
⇒2x – 4 = 0 ⇒ x = 2
Again differentiating w.r.t.x., we get
f″(x) = 2 > 0 (minimum)
Hence, f(x) is minimum at x = 2
Sol.18. (c)
Given function,
f(x) = |x–4|
Graph of f(x)



From graph, we observe that f(x) has minimum value at x = 4
Sol.19. (a)
Given functions,
$f(x) = x^3 + 2x^2 – 4x + 6$
and $f'(x) = 3x^2 + 4x – 4$
Now, for max or min of f(x), Put f′(x) = 0
$3x^2 + 4x – 4 = 0$
$\Rightarrow 3x^2 + 6x – 2x – 4 = 0$
⇒3x(x + 2) – 2 (x + 2) = 0
⇒(x + 2) (3x – 2) = 0
∴ f″(x) = 6x + 4
At x = – 2
f″(–2) = 6 (–2) + 4 = – 12 + 4
= – 8 < 0 (maximum)
So, max value of the f(x) exists at x = – 2.
Sol.20. (b)
Given function, $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$

$\Rightarrow f'(x) = \frac{\left[(x^2+x+1)(2x-1)-(x^2-x+1)(2x+1)\right]}{(x^2+x+1)^2}$

$\Rightarrow f'(x) = \frac{\begin{bmatrix} 2x^2+2x^2+2x+x^2-x-1-2x^3 \\ +2x^2-2x-x^2+x-1 \end{bmatrix}}{(x^2+x+1)^2}$

$\Rightarrow f'(x) = \frac{2x^2-2}{(x^2+x+1)^2}$

For maximum or minimum value of f(x),
Put f′(x) = 0
$\Rightarrow \frac{2(x^2-1)}{(x^2+x+1)} = 0$
$\Rightarrow x^2 – 1 = 0$
∴ x = 1 or – 1
Again,
$(x^2 + x +1) (4x) – (2x^2 – 2)$
$f''(x) = \frac{4(x^2+x+1)(2x+1)}{(x^2+x+1)^4}$

$= \dfrac{4(x^2+x+1)[x-(x^2-1)(2x+1)]}{(x^2+x+1)^4}$

$= \dfrac{4}{(x^2+x+1)^3} \times (x-2x^3+2x-x^2+1)$

$= \dfrac{4\times(-2x^3-x^2+3x+1)}{(x^2+x+1)^3}$

At x= 1,

$f''(1) = \dfrac{4(-2-1+3+1)}{(1+1+1)} = \dfrac{4}{27} > 0$ (minimum)

So, function f(x) is minimum at x = 1

At x = – 1

$f(-1) = \dfrac{4(2-1-3+1)}{(1-1+1)3}$

$= \dfrac{4\times(-1)}{(1)^2} = -4 < 0$ (maximum)

So, function f(x) is maximum at x = – 1.

Sol.21. (d)

Now, maximum value of the function.

At = x = –1, $f(-1) = \dfrac{1+1+1}{1-1+1} = 3$

Sol.22. (c)

We know that,

Slope of the tangent to the given curve = Slope of the given curve …(i)

Given curve, $y = e^{2x}$

On differentiating w.r.t.x., we get,

$\dfrac{dy}{dx} = 2e^{2x}$

$\Rightarrow \left(\dfrac{dy}{dx}\right)_{(0,1)} = 2.e^0 = 2 \times 1 = 2$

Hence, slope of the given curve = 2

Which also the slope of the tangent to the given curve at (0, 1)

Sol.23. (c)

Now, equation of tangent to the curve $y = e^{2x}$ at (0, 1) is

$(y-1) = \left(\dfrac{dy}{dx}\right)_{(0,1)} (x-0)$

$\Rightarrow (y-1) = 2(x-0)$

$\Rightarrow y - 1 = 2x$

$\therefore y = 2x + 1$

Since, the tangent meet X-axis

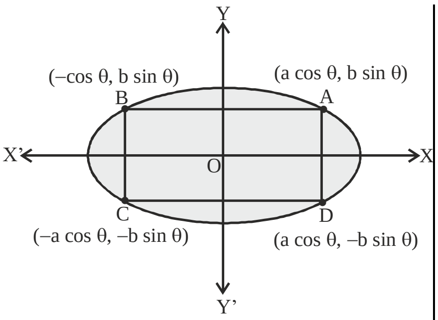
$\therefore$ Putting y = 0 in equation of tangent, we get

$0 = 2x + 1 \Rightarrow x = -1/2$

Hence, the tangent to the curve at (0,1) meet the X-axis at $\left(-\dfrac{1}{2}, 0\right)$

Sol.24. (b)

Let A(a cosθ, b sinθ) be any point on an ellipse. (Ist quadrant)



Coordinates of B = {a cos (π–θ), b sin (π–θ)}

=(–a cosθ, bsinθ) (IInd quadrant)

Coordinates of C={a cos(π+θ),b sin (π+θ)}

– (–a cosθ, – bsinθ) (III quadrant)

Coordinates of D={a cos(2π–θ), b sin(2π–θ)}

= (acosθ, –sinθ) (IV quadrant)

Now, area of rectangle ABCD=CD × AD

=(a cos θ + a cos θ) × (b sin θ + b sin θ)

= 2a cos θ × 2 b sin θ = 2 ab sin 2θ

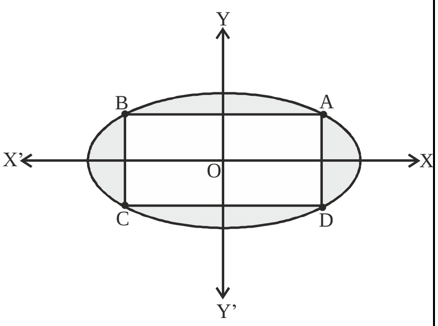
Here, area of rectangle will be greatest, when sin 2θ have its maximum value i.e. sin 2θ = 1

$\therefore$ Area of rectangle = 2ab × 1 = 2ab

Sol.25. (c)

We know that,

Area of the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is, Δ' = πab



$\therefore$ Required area = area of shaded region

= Area of an ellipse – Area of greatest rectangle

= πab – 2ab = ab (π–2)

Sol.26. (a)

Given curve,

$y = \sin^{-1}(\sin^2 x)$

On differentiating w.r.t.x., we get

$\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-(\sin^2 x)^2}} \cdot \dfrac{d}{dx}(\sin^2 x)$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{2\sin x.\cos x}{\sqrt{1-\sin^4 x}}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{\sin 2x}{\sqrt{1-\sin^4 x}}$

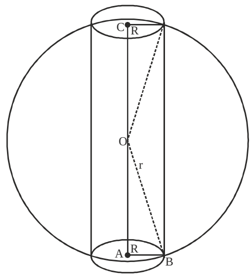
$\left(\dfrac{dy}{dx}\right)_{x=0} = \dfrac{\sin 0}{\sqrt{1-\sin 0}}$

$= \dfrac{0}{\sqrt{1-0}} = 0$

Sol.27. (a)

Let h be the height, R be the radius and V be the volume of cylinder

In ΔOAB, we have



$r^2 = R^2 + \left(\dfrac{h}{2}\right)^2$ …(i)

$\left(\because OA = \dfrac{h}{2} \text{ as } \Delta OAB \cong \Delta OCD\right)$

Clearly, $V = \pi R^2 h$

$\Rightarrow V(h) = \pi\left(r^2 - \dfrac{h^2}{4}\right)h$

[using equation (i)]

$\Rightarrow V(h) = \pi\left(r^2 h - \dfrac{h^3}{4}\right)$

$\Rightarrow V'(h) = \pi\left(r^2 - \dfrac{3h^2}{4}\right)$

…(ii)

For maximum, Put V′(h) = 0

$\Rightarrow r^2 = \dfrac{3h^2}{4} \Rightarrow h^2 = \dfrac{4r^2}{3}$

$\Rightarrow h = \dfrac{2r}{\sqrt{3}}$

($\because$ h > 0)

Again, differentiating equation (ii) w.r.t.h. we get

$V''(h) = \pi\left(\dfrac{-6\pi}{4}\right)$

$\Rightarrow V''\left(\dfrac{2r}{\sqrt{3}}\right) = \pi\left(\dfrac{-6h}{4}\right) < 0$

Thus, the volume is maximum when

$h = \dfrac{2r}{\sqrt{3}}$

Sol.28. (b)

Clearly, volume of cylinder is maximum when $h = \dfrac{2r}{\sqrt{3}}$

by using the relation $r^2 = R^2 + \left(\dfrac{h}{2}\right)^2$, we have

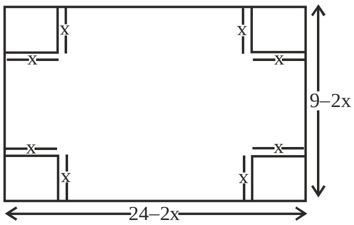
$R^2 = r^2 - \dfrac{h^2}{4} = r^2 - \dfrac{4r^2}{12} = \dfrac{8r^2}{12} = \dfrac{2r^2}{3}$

$\Rightarrow R = \sqrt{\dfrac{2r^2}{3}} = \dfrac{\sqrt{2}r}{\sqrt{3}}$

($\because$ R > 0)

Sol.29. (c)

Given, rectangular box is to be made from a sheet of size 24″ × 9″ by cutting out identical square of side length x from the four corners.



Clearly, the length of rectangular box = 24 – 2x and the width of rectangular box = 9 – 2x.
Let V be the volume of the box
$\therefore V = (24 - 2x) \cdot (9 - 2x) \cdot x$ ($\because$height of box = x inch)
$= (216 - 48x - 18x + 4x^2) \cdot x$
$V(x) = 4x^3 - 66x^2 + 216x$
$\Rightarrow V'(x) - 12x^2 - 132x + 216$
For maximum, put $V'(x) = 0$
$\Rightarrow 12x^2 - 132x + 216 = 0$
$\Rightarrow x^2 - 11x + 18 = 0$
$\Rightarrow (x - 9)(x - 2) = 0$
$\Rightarrow x = 9$ or $x = 2$
Now, $V''(x) = 24x - 132$
$\therefore V''(9) = 216 - 132 = -84 > 0$
and $V''(2) = 48 - 132 = -84 < 0$
Thus, volume is maximum when
x = 2 inch.

Sol.30. (a)
Volume of box = (24 – 4) (9 – 4). 2 = 20 × 5 × 2 = 200 cu inch

Sol.31. (c)
We have,
$f(x) = 0.75x^4 - x^3 - x^3 - 9x^2 + 7$
$\Rightarrow f'(x) = 3x^3 - 3x^2 - 18x$
$\Rightarrow f'(x) = 3x(x^2 - x - 6) = 3x(x - 3)(x + 2)$
For critical points, put $f'(x) = 0$
$\Rightarrow 3x + (x - 3)(x + 2) = 0$
$\Rightarrow x = 0, x = 3$ or $x = -2$
Thus, there are only three critical points which could possibly be the point of local maxima or local minima.
Clearly, –2 and 3 are the point of local minima and 0 is the point of local maxima.
$\therefore$The maximum value of the function is given by
f(0) = 7

Sol.32. (c)
Hence, both statements are true.

Sol.33. (b)
$f(x) = \dfrac{x^2 - 1}{x^2 + 1}$
$f'(x) = 2x[(x^4 - 1)\log(x^2 + 1)]$
Putting $f'(x) = 0$, x = 0 or x = 1
Clearly, at x = 0 it has minima

Sol.34. (c)
and minimum value is -1 at x = 0

Sol.35. (c)
I.Let $f(x) = \dfrac{e^x + e^{-x}}{2}$
$\Rightarrow f'(x) = \dfrac{e^x - e^{-x}}{2}$
$= \dfrac{1}{2}\left(e^x - \dfrac{1}{e^x}\right) = \dfrac{1}{2}\left(\dfrac{e^{2x} - 1}{e^x}\right)$ …(i)
Now, for $x \geq 0$, we have $2x \geq 0$
$= e^{2x} \geq e^0$ ($\therefore e^x$ is an increasing function) $= e^{2x} \geq 1$
also for $x \geq 0 \Rightarrow e^x \geq 1$
$\therefore$from equation (i), we have $f'(x) = \dfrac{1}{2}\left(\dfrac{e^{2x} - 1}{e^x}\right) \geq 0$
So, f(x) is increasing function on $(0, \infty)$
II. Let g(x)
$= \dfrac{e^x - e^{-x}}{2} \Rightarrow g'(x) = \dfrac{e^x + e^{-x}}{2} > 0$
[$\because e^x$ and e^{-x} both are greater than zero in $(-\infty, \infty)$]
So, g(x) is an increasing function on $(-\infty, \infty)$.
Hence, both the statements are true.

Sol.36. (c)
$y = \left(\dfrac{1}{x}\right)^{2x^2}$
Taking log of both sides
$\log y = 2x^2 \log\left(\dfrac{1}{x}\right)$
$= 2x^2 \cdot (\log 1 - \log x) = -2x^2 \cdot \text{Log}x$
Differentiating w.r.t.x.,
$\dfrac{1}{y}\dfrac{dy}{dx} = -\dfrac{2x^2}{x} - \log x \times 4x$
or $\dfrac{dy}{dx} = y[-2x - 4x.\log x]$
Putting $\dfrac{dy}{dx} = 0$, we get
$y[-2x - 4x \log x] = 0$
or $-2x = 4x \log x \Rightarrow \log_e x = -\dfrac{1}{2}$
or $\left(x = \dfrac{1}{\sqrt{e}}\right)$
$\therefore$the function has the maximum value at $x = \dfrac{1}{\sqrt{e}}$

Sol.37. (c)
Putting the value of x in given function
$y = \left(\dfrac{1}{\dfrac{1}{\sqrt{e}}}\right)^{\frac{2}{3}} = \sqrt{e}^{2/e} = (e^{1/e})$

Sol.38. (d)
$f(x) = -2x^3 - 9x^2 - 12x + 1$
$f'(x) = -6x^2 - 18x - 12$
$f'(x) = -6(x^2 + 3x + 2) = -6(x+2)(x+1)$
Clearly, f(x) is decreasing
When $x \in (-\infty, -2) \cup (-1, \infty)$

Sol.39. (a)
If, f(x) is increasing then $f'(x) > 0$
or $-6(x + 2)(x + 1) > 0$
or $(x + 2)(x + 1) < 0 \Rightarrow -2 < x < -1$

Sol.40. (a)
I.Function $f(x) = x^2 + 2\cos x$, is increasing in the interval $(0, \pi)$
$f(x) = 2x - 2\sin x \Rightarrow 2(x - \sin x)$
For interval increasing $f'(x) > 0$, $x \in (0, \pi)$
$2(x - \sin x) > 0$
So, for every $x \in (0, \pi)$, $f'(x) > 0$
Hence, f(x) is increasing in interval $(0, \pi)$
II.$f'(x) = \log\left(\sqrt{1 + x^2} - x\right)$
$\Rightarrow f'(x) = \left(\dfrac{1}{\sqrt{1 + x^2} - x}\right) \times \left(\dfrac{2x}{2\sqrt{1 + x^2}} - 1\right)$
$= \dfrac{\sqrt{1 + x^2} + x}{\left(1 + x^2\right) - x^2} \times \dfrac{x - \sqrt{1 + x^2}}{1 + x^2}$
$= \dfrac{x^2 - \left(1 + x^2\right)}{\sqrt{1 + x^2}} = \dfrac{-1}{\sqrt{1 + x^2}}$
Here, $f'(x)$ decreases from $-\infty$ to 0
But increase from 0 to ∞
Thus, only statement (I) is correct.

Sol.41. (c)
1. y = lnx
dy/dx = 1/x it is positive for $x \in (0, \infty)$
first statement is correct
2. $y = e^x - x(\ln x)$
$dy/dx = e^x - 1 - \ln x$ it is positive for $(1, \infty)$
second statement is also correct

Sol.42. (c)
$f(x) = \dfrac{x^2}{e^x}$
$f'(x) = \dfrac{e^x.2x - x^2.e^x}{e^{2x}} = \dfrac{e^x.x(2 - x)}{e^{2x}}$
For monotonically increasing $f'(x) > 0$
$\dfrac{e^x.x(2 - x)}{e^{2x}} > 0$
$2 - x > 0$
$x < 2$
$x > 0$
So, $0 < x < 2$
Hence, f(x) is monotonically increasing if 0 < x < 2.

Sol.43. (c)
$f(x) = (x-1)^2(x+1)(x-2)^3$
$f'(x) = 2(x-1)(x+1)(x-2)^3 + (x-1)^2(x-2)^3 + 3(x-1)^2(x+1)(x-2)^2$
$= (x-1)(x-2)^2[2(x+1)(x-2) + (x-1)(x-2) + 3(x^2-1)]$
$= (x-1)(x-2)^2[2x^2 - 2x - 4 + x^2 - 3x + 2 + 3x^2 - 3]$
$= (x-1)(x-2)^2[6x^2 - 5x - -5] = 0$
$x = 2, 1, \dfrac{5 - \sqrt{145}}{12}, \dfrac{5 + \sqrt{145}}{12}$
now $f''(x) = 2(x-2)(3x-5)(5x^2-5x-1)$
for x = 2, $f''(x) = 0$
for x = 1, $f''(x) < 0$
hence x = 1 has local maxima
$x = \dfrac{5 \pm \sqrt{145}}{12}, f''(x) > 0$
hence, $\dfrac{5 \pm \sqrt{145}}{12}$
has local minima

Sol.44. (b)
Check above solution

Sol.45. (b)
$f(x) = \begin{cases} 1 - x + x^2, & x < 1 \\ -1 + x + x^2, & x \geq 1 \end{cases}$
$f'(x) = \begin{cases} 2x - 1, x < 1 \\ 2x + 1, x \geq 1 \end{cases}$
Case I. x < 1
$f'(x) > 0$
$\Rightarrow 2x - 1 > 0$
$\Rightarrow x > \dfrac{1}{2}$
$\therefore f(x)$ is increasing $\left[\dfrac{1}{2}, \infty\right]$
Case II. $x \geq 1$
$\Rightarrow f'(x) > 0$
$\Rightarrow 2x + 1 \geq 1 \Rightarrow x > -1/2$

$\therefore$ f(x) is decreasing $(-\infty, \frac{1}{2})$

Sol.46. (c)
Clearly, the function has local minima at only one point in $(-\infty, \infty)$

Sol.47. (d)
$f(\theta) = 4(\sin^2\theta + \cos^4\theta)$
$f'(\theta) = 4(2\sin\theta\cos\theta - 4\cos^3\theta\sin\theta)$
$= 8\sin\theta\cos\theta(1 - 2\cos^2\theta) = 0$
$= \theta = 0^0, 90^0, 45^0$
max at $\theta = 0^0, 90^0$ is 4

Sol.48. (d)
By above solution
min at $x = 45^0$ is 3

Sol.49. (c)
$3 \le 4(\sin^2\theta + \cos^4\theta) \le 4$
$f(\theta)$ can not be 2 but $f(\theta)$ can be 7/2

Sol.50. (b)
$$f(x) = \frac{27\left(x^{2/3} - x\right)}{4} = 1$$
Put $x^{1/3} = y$
$x = y^3$
$y^2 - y^3 = 4/27$
$\Rightarrow y3 - y2 + 427 = 0$
$\Rightarrow 27y^3 - 27y^2 + 4 = 0$
put $y = -1/3$
then $(3y+1) = 0$ is a factor of cubic equation
$(3y+1)(9y^2 - 12y + 4) = 0$
$(3y+1)(3y-2)^2 = 0$
Hence $y = -1/3, 2/3$
Thus f(x) = 1 has two solutions.

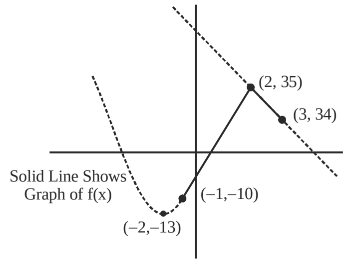
Sol.51. (a)
same as above solution f(x) = - 1 has only one solutions.

Sol.52. (b)
$k\sin x + \cos 2x = 2k - 7$
$k\sin x + 1 - 2\sin^2 x = 2k - 7$
$2\sin^2 x - k\sin x + (2k - 8) = 0$
$$\sin x = \frac{k \pm \sqrt{k^2 - 8(2k-8)}}{4}$$
$$\sin x = \frac{k \pm \sqrt{(k-8)^2}}{4}$$
$$\sin x = \frac{k \pm (k-8)}{4}$$
$$\sin x = 2, \frac{k-4}{2}$$
For real solution
$$-1 \le \frac{k-4}{2} \le 1$$
$$2 \le k \le 6$$
Minimum value of k = 2

Sol.53. (d)
By above solution maximum value of k = 6

Sol.54. (b)
$y = x^3\sin x$
$y' = 3x^2\sin x + x^3\cos x$
$= x^2(3\sin x + x\cos x) = 0$ than $x = 0$
$y'' = 6x\sin x + 3x^2\cos x - 3x^2\cos x - x^3\sin x = 0$
$y''' = 6x\cos x + 6\sin x - 3x^2\sin x - x^3\cos x = 0$
$y'''' = 6\cos x - 6x\sin x + 6\cos x - 6x\sin x - 3x^2\cos x - 3x^2\cos x + x^3\sin x = 12$ at $x = 0$
local minimum at x = 0

Solutions (for next two)
Given, f(x)
$$= \begin{cases} 3x^2 + 12x - 1, & -1 \le x \le 2 \\ 37 - x, & 2 \le x \le 3 \end{cases}$$
$\because 3x^2 + 12x - 1 = 3(x^2 + 4x + 4) - 12 - 1 = 3$
$(x+2)^2 - 13$
Now, we can draw graph of f(x) in which (–2, –13), is vertex of $3x^2 + 12x - 1$



Sol.55. (c)
From graph, f(x) > 0 for $x \in [-1, 2]$
$\therefore$ f(x) is increasing in the interval [–1,2]
and f(x) < 0 for $x \in (2, 3)$
$\therefore$ f(x) is decreasing in the interval (2,3)

Sol.56. (a)
$\because$ Graph of f(x) does not break at x = 2.
$\therefore$ f(x) is continuous at x = 2
$\because$ From graph, f(x) attains maximum value at x = 2, which is 35. Since, graph of f(x) makes sharp edge at x = 2
$\therefore$ f(x) is not differentiable at x = 2

Sol.57. (b)
Given, $f(x) = \begin{cases} \frac{e^x - 1}{x}, & x > 0 \\ 0, & x = 0 \end{cases}$
$$\Rightarrow f'(x) = \frac{x\frac{d}{dx}\left(e^x - 1\right) - \left(e^x - 1\right)\frac{d}{dx}(x)}{x^2}$$
$$\Rightarrow f'(x) = \frac{xe^x - e^x + 1}{x^2}$$
$$\Rightarrow f'(x) = \frac{e^x(x-1) + 1}{x^2}$$
Clearly, $f'(x) > 0$, $x > 0$
$\therefore$ f(x) is a strictly increasing function in (0,x)

Sol.58. (d)
I. For right continuity at x = 0, $\lim\limits_{x \to 0^+} f(x) = f(0)$
Now, $\lim\limits_{x \to 0^+} f(x) = \lim\limits_{x \to 0^+} f(0 + h)$
$= \lim\limits_{h \to 0} f(h) = \lim\limits_{h \to 0} \frac{e^h - 1}{h} = 1$
and f(0) = 0
So, $\lim\limits_{x \to 0^+} f(x) \ne f(0)$
$\Rightarrow$ f(x) is not right continuous at x = 0
II. At x = 1,
$$\lim_{x \to 1^-} = \lim_{h \to 0} \frac{e^{(1-h)} - 1}{(1-h)} = \frac{e - 1}{1} = e - 1$$
and $f(1) = \frac{e^1 - 1}{1} = e - 1$
$\therefore$O $\lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^+} f(x) = f(1)$
$\Rightarrow$ f(x) is continuous at x = 1

Sol.59. (c)
We know that the area of a triangle inscribed in a circle is maximum when the triangle is equilateral triangle.
Also, Radius of circum circle
$$= \frac{\text{Side of triangle}}{\sqrt{3}}$$
or $\sqrt{3}a$ = side
$\therefore$ Area of triangle =
$$\frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}\left(\sqrt{3}a\right)^2 = \frac{3\sqrt{3}a^2}{4}$$

Sol.60. (a)
$f(x) = 3\sin x - 4\sin^3 x = \sin 3x$
We know that,
$-1 \le \sin 3x \le 1$
$\Rightarrow -\frac{\pi}{2} \le 3x \le \frac{\pi}{2}$
$\Rightarrow -\frac{\pi}{6} \le x \le \frac{\pi}{6}$ $\therefore$ The range or the interval =
$$\frac{\pi}{6} - \left(-\frac{\pi}{6}\right)$$
$$= \frac{\pi}{3}\left\{\begin{matrix} \because \text{Range} = b - a \\ \text{if } a \le x \le b \end{matrix}\right\}$$

Sol.61. (a)
max value of $\sin^2 x$ is 1
max value of $4\sin^2 x$ is 4
max value of $4\sin^2 x + 1$ is 5

Sol.62. (c)
$$\Rightarrow y = x + \frac{1}{x}$$
$$\Rightarrow y' = 1 - \frac{1}{x^2} = 0$$
For increasing y' > 0
$$\Rightarrow y' = \frac{x^2 - 1}{x^2} > 0$$
$$\Rightarrow (x+1)(x-1) > 0$$
Function is increasing for x < -1 or x > 1
And decreasing in (-1,1)

Sol.63. (a)
$f(x) = x(x^2 - 1)$
or $f'(x) = x(2x) + (x^2 - 1) = 3x^2 - 1$
or $f''(x) = 6x$
At $f'(x) = 0 \Rightarrow x \,\square\, \frac{1}{\sqrt{3}}$
$$f(x)_{max} = f\left(\frac{-1}{\sqrt{3}}\right) = \frac{-1}{\sqrt{3}}\left(\frac{1}{3} - 1\right) = \frac{2}{3\sqrt{3}}$$
$$f(x)_{min} = f\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}}\left(\frac{1}{3} - 1\right) = \frac{-2}{3\sqrt{3}}$$

Sol.64. (b)
Since the maximum value of a sin x + b sin x = $\sqrt{a^2 + b^2}$
$\therefore$ The maximum value of sin x + cos x = $\sqrt{1+1} = \sqrt{2}$
The maximum value of 3sin x + 4 cos x = $\sqrt{3^2 + 4^2} = 5$
The maximum value of 2 sin x + cos x = $\sqrt{2^2 + 12} = \sqrt{5}$
The maximum value of sin x = 3 cos x = $\sqrt{1^2 + 3^2} = \sqrt{10}$

Sol.65. (b)
$$y = \frac{\log_e x}{x}$$
$$\frac{dy}{dx} = \frac{x.\frac{1}{x} - \log_e x.1}{x^2} = \frac{1 - \log_e x}{x^2}$$

Putting $\frac{dy}{dx} = 0$

$\frac{1-\log_e x}{x^2} = 0$

or $1 - \log_e x = 0$

or $\log_e x = 1$

$x = e$

$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x^2} - \frac{\log_e x}{x^2}\right)$

$= \frac{-2}{x^3} - \frac{1}{x^2} \times \frac{1}{x} - \log_e x\left(\frac{-2}{x^3}\right)$

$= -\frac{2}{x^3} - \frac{1}{x^3} + \frac{2.\log_e x}{x^3}$

$= \frac{2\log_e x}{x^3} - \frac{3}{x^3}$

Putting x= e, in the above,

$\frac{d^2y}{dx^2} = \frac{2}{e^3} - \frac{3}{e^3} = -\frac{1}{e^3} = -ve$

∴The given expression has maximum value at x = e

Thus, the required value = $\frac{\log_e e}{e} = \frac{1}{e}$

Sol.66. (b)

The surface area (given) = $2\pi rh + \pi r^2$ (without lid)

or $S = 2\pi rh + \pi r^2$

or $h = \frac{S - \pi r^2}{2\pi r}$ …(i)

Now, volume, $v = \pi r^2 h$

or $v = \pi r^2 \left(\frac{S - \pi r^2}{2\pi r}\right)$

[from (i)]or $v = \frac{S}{2} r - \frac{\pi}{2} r^3$

$\Rightarrow \frac{dv}{dr} = \frac{5}{2} - \frac{\pi}{2} \times 3r^2$

If the volume is maximum, then the critical points are given

as, $\frac{dv}{dr} = 0$

or $\frac{S}{2} - \frac{\pi}{2} \times 3r^2 = 0$

or $S = 3\pi r^2$

or $2\pi rh + \pi r^2 = 3\pi r^2$ …(ii)

∴The ratio of diameter to height is given as 2r : h = 2r : r [From (ii)]

= 2 : 1

Thus, the diameter is 2 times that of the height.

Sol.67. (b)

F(x) = x – sinx

X is always greater than sinx for all x> 0

f'(x) = 1 – cosx >0 so function is always increasing.

Sol.68. (c)

maximum value of $y = 16\sin\theta - 12\sin^2\theta$

$\frac{dy}{d\theta} = 16\cos\theta - 24\sin\theta\cos\theta = 0$

$8\cos\theta(2 - 3\sin\theta) = 0$

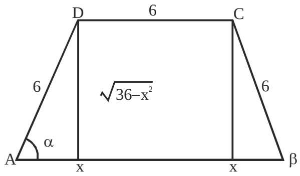
$\sin\theta = 2/3$

$y = 16\left(\frac{2}{3}\right) - 12\left(\frac{2}{3}\right)^2 = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$

Sol.69. (c)

A = area = $\frac{1}{2}(6+6+2x)\sqrt{36-x^2}$

$= (6 + x)\sqrt{36-x^2}$



$\frac{dA}{dx} = (6+x)\left(\frac{-2x}{2\sqrt{36-x^2}}\right) + \sqrt{36-x^2}$

$= \sqrt{36-x^2} - \frac{x(6+x)}{\sqrt{36-x^2}} = \frac{36-6x-2x^2}{\sqrt{36-x^2}}$

$\frac{dA}{dx} = 0 \Rightarrow 36 - 6x - 2x^2 = 0 \Rightarrow x = 3$

$\frac{d^2A}{dx^2} < 0$

Now, $\cos\alpha = \frac{x}{6} = \frac{3}{6} = \frac{1}{2} \Rightarrow \alpha = \frac{\pi}{3}$

Sol.70. (d)

Fourth side = 6 + 6 = 12

Sol.71. (c)

Maximum area = $9 \times 3\sqrt{3} = 27\sqrt{3}$

Sol.72. (a)

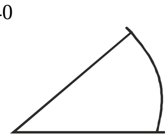
$[x(x-1) + 1]^{1/3}$ is minimum when $x^2 - x + 1$ is minimum, $0 \le x \le 1$

Minimum value of $x^2 - x + 1$ is 3/4

⇒Required value = $\left(\frac{3}{4}\right)^{1/3}$

Sol.73. (c)

$\ell + 2r = 40$



Area = $\frac{r^2(\text{angle})}{2} = \frac{r^2\left(\frac{L}{r}\right)}{2} = \frac{Lr}{2}$

$= \frac{(40-2r)r}{2} = \frac{40r - 2r^2}{2}$

for max area $\frac{dA}{dr} = \frac{40-4r}{2} = 0$

$\Rightarrow r = 10$

Sol.74. (a)

$f(x) = x^2 - 5x + 6$

$\Rightarrow f'(x) = 2x - 5$

$\Rightarrow f'(x) < 0$

$\Rightarrow 2x - 5 < 0$

$\Rightarrow x < 5/2$

Sol.75. (a)

$f(x) = x\sin x + \cos x + 1/2\cos^2 x$

$\Rightarrow f'(x) = x\cos x + \sin x - \sin x - \sin x\cos x$

$= \cos x(x - \sin x) > 0$ in $\left(0, \frac{\pi}{2}\right)$

Sol.76. (b)

$y = \cos^{-1}(\sin x) = \cos^{-1}\cos\left(\frac{\pi}{2} - x\right) = \frac{\pi}{2} - x$

$dy/dx = \tan\theta = -1 \Rightarrow \theta = \frac{3\pi}{4}$

Sol.77. (b)

Let $Z = a^2x + b^2y$ and $xy = c^2 \Rightarrow y = \frac{c^2}{x}$ $Z = a^2x + b^2\frac{c^2}{x}$

$\frac{dz}{dx} = a^2 - \frac{b^2c^2}{x^2} = 0$

$x^2 = \frac{b^2c^2}{a^2} \Rightarrow x = \frac{bc}{a}$

$Z = a^2\frac{bc}{a} + b^2\frac{c^2}{\frac{bc}{a}} = abc + abc = 2abc$

Sol.78. (c)

The minimum distance from the point (4,2) to $y^2 = 8x$ is equal to

Let any point on the curve (h,k) that is at minimum distance from given point

$D = \sqrt{(h-4)^2 + (k-2)^2}$

it will minimum when $(h-4)^2 + (k-2)^2$ is minimum

$(h-4)^2 + (k-2)^2 = h^2 - 8h + 16 + k^2 - 4k + 4$

and $k^2 = 8h$(point is on the curve)

$\left(\frac{k^2}{4}\right)^2 - k^2 + 16 + k^2 - 4k + 4 = \frac{k^4}{64} - 4k + 20$

derivative $\frac{k^3}{16} - 4 = 0$

k = 4 than h = 2

so nearest point is (2,4)

(2,4) and smallest distance is distance between (4,2) = $2\sqrt{2}$

Sol.79. (a)

$V = \frac{1}{2}\pi r^2 h \Rightarrow h = \frac{2V}{\pi r^2}$

$A = h(2r) + \pi r^2 + \pi rh$

$A = h(2r) + \pi r^2 + \pi rh$

$A = \frac{2V}{\pi r^2}(2r) + \pi r^2 + \pi r\frac{2V}{\pi r^2}$

$A = \frac{4V}{\pi r} + \pi r^2 + \frac{2V}{r}$

$\frac{dA}{dr} = -\frac{4V}{\pi r^2} + 2r^2 - \frac{2V}{r^2} = 0$

$r^3 = \frac{V(2+\pi)}{\pi^2}$

$\frac{h}{2r} = \frac{\frac{2V}{\pi r^2}}{2r} = \frac{2V}{\pi r^2} = \frac{2V}{\pi\frac{V(2+\pi)}{\pi^2}} = \frac{\pi}{\pi+2}$

Sol.80. (c)

Let roots of $2x^2 - 2(k-2)x - (k+1) = 0$ are α and β

$\alpha + \beta = k - 2$, and $\alpha\beta = -(k+1)/2$

$z = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$z = \alpha^2 + \beta^2 = (k-2)^2 + (k+1)$

$z = k^2 - 3k + 5$

$z' = 2k - 3 = 0$

so k = 3/2

z'' = - 3 so at k = 3/2 there will be a minimum value.

Sol.81. (b)

at y axis x = 0 then P (0,m)
y = me^{mx}
y' = m^2e^{mx} ....(i)
put coordinates of P in (i)
y ' = m^2
Sol.82. (c)
slope = tanθ = m^2
θ = $\tan^{-1}(m^2)$
angle with y axis = $90^0 - \tan^{-1}(m^2)$
= $\cot^{-1}(m^2)$
= $\sin^{-1}\left(\frac{1}{\sqrt{1+m^4}}\right)$
Sol.83. (d)
equation of tangent
y − m = m^2 (x−0)
y = m^2x + m
Sol.84. (a)
$$f(x)=\frac{x^3}{3}-\frac{5x^2}{2}+6x+7$$
$$f(x)=\frac{x^3}{3}-\frac{5x^2}{2}+6x+7$$
f '(x) = $x^2 - 5x + 6$
for increasing $x^2 - 5x + 6 > 0$
(x−2)(x−3) > 0
increases in T = (−∞,2)∪(3,∞)
decreasing in S = (2,3)
Sol.85. (a)
f(x) = $3x^4 - 20x^3 - 12x^2 + 288x + 1$
f '(x) = $12x^3 - 60x^2 - 24x + 288$
f '(x) = $12(x^3 - 5x^2 - 2x + 24)$
f '(x) = 12(x+2)(x−3)(x−4)
increasing in (−2,3) ∪(4,∞)
Sol.86. (b)
increasing in (−∞,−2) ∪(3,4)
Sol.87. (c)
y = sinx.cosx
y = 1/2.2 sinx.cosx
y = 1/2 sin2x
maximum value of sin2x is 1
so maximum value of 1/2 sin2x is 1/2
Sol.88. (a)
minimum value of cosθ is −1
so minimum value of 3cosθ will be −3.
Sol.89. (c)
By graphs of logx and tanx we can check that both statements are correct.
Sol.90. (d)
given that
l + b = k ...(i)
l = k −b
Area = A = l.b
A = (k−b)b = kb−b^2
diff.w.r.t. x
$\frac{dA}{db}=k-2b=0$ ⇒ b = k/2
then l = k/2 by equation (i)
max area = k/2 . k/2 = $k^2/4$
Sol.91. (c)
f(10.1) − f(10)
= $[3(10.1)^2 + 2] - [3(10)^2 + 2] = 6.03$
Sol.92. (a)
Minimum value of |x – 1| exists at x =1 that is zero.
Sol.93. (a)
given that $\frac{dr}{dt}=0.7cm/\sec$
C = 2πr
$$\frac{dC}{dt}=2\pi\frac{dr}{dt}=2\times\frac{22}{7}\times0.7=4.4cm/\text{s}$$
Sol.94. (b)
By graph of sinx we can see that
1. sinx is increasing in $\left(0,\frac{\pi}{2}\right)$ not in $(0,\pi)$
2. sinx is decreasing in $\left(\frac{5\pi}{2},3\pi\right)$
only 2nd statement is correct
Sol.95. (a)
Area of a triangle will be maximum for given perimeter when that is an equilateral triangle.
perimeter = 24
side = 8
altitude = $\frac{\sqrt{3}}{2}(8)=4\sqrt{3}$ cm
Sol.96. (c)
y = $-x^3 + 3x^2 + 2x - 27$
slope = m = $\frac{dy}{dt}=-3x^2+6x+2$
for maximum slope
$$\frac{dm}{dx}=-6x+6=0$$
x = 1
Sol.97. (a)
given that
x + y = 20 ...(i)
x = 20 − y
P = xy
P = (20−y)y = 20y−y^2
diff.w.r.t. x $\frac{dP}{db}=20-2y=0$ ⇒ y = 10
then x = 10 by equation (i)
maximum value of P = 10×10 = 100
Sol.98. (a)
y = sin2x.cos2x
y = 1/2.2 sin2x.cos2x
y = 1/2 sin4x maximum value of sin4x is 1
so maximum value of 1/2 sin4x is 1/2
Sol.99. (b)
y = $x^3 + x^2$ + kx, $\frac{dy}{dx}$ =$3x^2$ + 2z + k
No extreme value
So $\frac{dy}{dx}$ never be equal to zero
$b^2 - 4ac < 0$
⇒4 – 4(3) (k) < 0
⇒4 < 12 k
⇒3k > 1
Sol.100. (a)
f(x) = x^2 – kx
f'(x) = 2x – k ≥ 0
2x ≥ k
⇒ 1 ≤ x ≤ ∞
⇒ 2 ≤ 2x ≤ ∞
2x is greater than or equal to 2.
But k < 2x
So k < 2
Sol.101. (d)
x^2 = y, $\frac{dy}{dx}=2x$
at (1, 1), $\frac{dy}{dx}=2$
tan θ = 2, Then $\frac{\pi}{3}<\theta<\frac{\pi}{2}$
Sol.102. (a)
y=$\cos^{-1}$ (cos x)
y=π–x $\frac{dy}{dx}=-1$
true x ∈ [0, π], y=π – cos–1 (cos x)
y=π–x
Sol.103. (d)
f(x) = x^2 + 1
f'(x) = 2x > 0 for all positive x in internal (1, 2)
function is increasing
minimum value will attain at n = 1
and maximum value at x = 2
f(1) = 2, f(2) = 5
But 1 and 5 cannot put at x we due to open internal. Both statements are incorrect.
Sol.104. (b)
y=e^x $\frac{dy}{dx}=e^x$
$\left(\frac{dy}{dx}\right)_{(0,1)}=e^0=1$
Equation of tangent at (0, 1)
y–1 = 1 (x – 0)
y– x = x
x–y + 1 = 0
this tangent will cut x axis at (–1, 0)
Sol.105. (b)
For maximum value
$\frac{d}{d\theta}$ (sinθ + sinθcosθ) = 0
cosθ – $\sin^2\theta$ + $\cos^2\theta$ = 0
2 $\cos^2\theta$ + cosθ –1 = 0
$2\cos^2$ + 2cosθ – cosθ –1 = 0
2cosθ (cosθ+1)–1(cosθ+1)= 0
(cosθ+1) (2cosθ–1) = 0
cosθ = $\frac{1}{2}$, θ = $\frac{\pi}{3}$
Sol.106. (a)
$$\Rightarrow y=x+\frac{1}{x}\ \Rightarrow y'=1-\frac{1}{x^2}=0$$
$$\Rightarrow x=1,-1$$
$$\Rightarrow y''=\frac{2}{x^3}$$
y’’ > 0 for x = 1 so x = 1 have local minima and at x= 1 minimum value is 2
y’’ < 0 for x = -1 so x = -1 have local maxima and at x= -1 maximum value is
- 2.
Only statement 1 is correct.
Sol.107. (c)
Maximum area of rectangle will obtained when it’s a square shape
Diameter of circle = diagonal of square
$2R=\sqrt{2}a=4$
$a=2\sqrt{2}$
Area of square =a^2 = 8 square unit
Sol.108. (d)
Let p =a , q = a + 2d and a + 4d = 3
Product M =pq
M = a(a+2d)
M=(3-4d)(3-4d+2d)
M=(3-4d)(3-2d)
M = 9 – 18d + 8d²
M’ = 16d – 18 = 0

d = 9/8

Sol.109. (b)

y = sin4x +2x

$\frac{dy}{dx}$ =4cos 4x + 2 = 0

cos4x = $-\frac{1}{2}$

cos α = $-\frac{1}{2} \Rightarrow \alpha = \frac{2\pi}{3}$

4x = 2xπ ± $\frac{2\pi}{3}$

x= $\frac{n\pi}{2} \pm \frac{\pi}{6}$

put n = 0 $\Rightarrow$ x = $\frac{\pi}{6}$

put n=1 $\Rightarrow$ = $\frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$

only 2 values of x.

Sol.110. (b)

$\frac{1}{\tan x + \cot x}$ will take its maximum value when tanx + cotx takes minimum value

And we know that minimum value of a tanθ + b cot θ is $2\sqrt{ab}$

So minimum value of tanx + cotx is 2

Then maximum value of $\frac{1}{\tan x + \cot x}$ is $\frac{1}{2}$

Sol.111. (b)

f(x) = $\frac{x^3}{3} - \frac{7x^2}{2} + 6x + 5$

for decreasing f'(x) < 0

$x^2 - 7x + 6 < 0$

(x – 1) (x– 6) < 0

x∈ (1, 6)

Sol.112. (a)

Xy = 4225

z =x + y

z = x + $\frac{4225}{x}$

$\frac{dz}{dx} = 1 - \frac{4225}{x^2} = 0$

x=65

then y = 65

x + y = 65 + 65 = 130

Sol.113. (a)

f(x) = $p^2 \sec^2 x + q^2 \operatorname{cosec}^2 x$

f'(x) = p^2 .2secx.secxtanx - q^2 cosecx.cosecxcotx = 0

$p^2\sec^2 x\tan x = q^2\operatorname{cosec}^2 x\cot x$

$$\frac{p^2 \sin x}{\cos^3 x} = \frac{q^2 \cos x}{\sin^3 x}$$

$$\frac{\sin^4 x}{\cos^4 x} = \frac{q^2}{p^2}$$

$$\tan^2 x = \frac{q}{p}$$

Sol.114. (b)

f(x) = $\sum_{j=1}^{7}(x - j)^2$

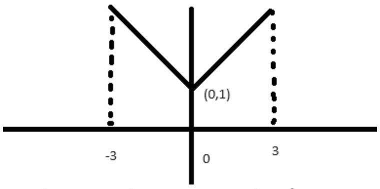
f(x) = $(x – 1)^2 + (x – 2)^2 + (x – 3)^2 + (x – 4)^2 + (x – 5)^2 + (x – 6)^2 + (x – 7)^2$

f'(x) = 2(x – 1)+ 2(x – 2)+ 2(x – 3)+ 2(x – 4)+ 2(x – 5)+ 2(x – 6)+ 2(x – 7)

f'(x) = 14x – 56 = 0

x = 4

Sol.115. (b)



By above graph we can say that function contain maximum value at 3 and -3 , minimum value at x = 0

Sol.116. (b)

let y = x^2 + 2x + 11

y = $(x + 1)^2$ + 10

minimum value of y is 10.

So minimum value of f(x) = $\log_{10}(x^2 + 2x + 11)$ = $\log_{10} y = \log_{10}10 = 1$

Sol.117. (a)

1. f(x) = *l*nx

f'(x) = 1/x > 0 for all x > 0

f(x) is increasing

2. g(x) = $e^x + e^{-x}$

g'(x) = $e^x – e^{-x}$ > 0 for all x > 0

g(x) is increasing

Sol.118. (d)

m(θ) = $\cot^2\theta + n^2\tan^2\theta + 2n$

$$\frac{dm}{d\theta} = 2\cot\theta.(-\operatorname{cosec}^2\theta) + n^2 2\tan\theta.\sec^2\theta$$

$$\frac{dm}{d\theta} = -\frac{2\cos\theta}{\sin^3\theta} + \frac{n^2 2\sin\theta}{\cos^3\theta} = 0$$

$$\frac{\sin^4\theta}{\cos^4\theta} = \frac{1}{n^2} = \tan^4\theta$$

$$\cot^2\theta = n$$

Minimum value of function exists for

$$\cot^2\theta = n$$

Minimum value of

m(θ) = $\cot^2\theta + n^2\tan^2\theta + 2n$

m(θ) = n + n + 2n = 4n

Sol.119. (b)

By above solution

$$\cot^2\theta = n$$

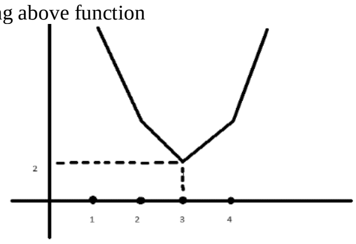
Sol.120. (b)

f(x) = |x–2|+ |3–x| + |4–x|,

f(x) = |x–2|+ |x–3| + |x–4|,

$$f(x) = \begin{cases} -3x+9 & x<2 \\ -x+1 & 2<x<3 \\ x-1 & ,3<x<4 \\ 3x-9 & ,x>4 \end{cases}$$

Using above function



At x = 3 function attain minimum value

Sol.121. (a)

At x = 3 function attain minimum value

f(3) = |3–2|+ |3–3| + |3–4|= 2

Sol.122. (b)

$$x = \frac{\sin^2 A + \sin A + 1}{\sin A}$$

$$x = \sin A + 1 + \frac{1}{\sin A} = \sin A + \frac{1}{\sin A} + 1$$

We know that

Minimum value of $n + \frac{1}{n}$ exists at n = 1 that is 2

So minimum value of $\sin A + \frac{1}{\sin A}$ is 2

than minimum value of $\sin A + \frac{1}{\sin A} + 1$ is 3

Sol.123. (d)

Minimum value of $n + \frac{1}{n}$ exists at n = 1 that is 2

So SinA = 1 and $A = \frac{\pi}{2}$

Sol.124. (a)

A function is defined by

f(x)= $\begin{vmatrix} x+1 & 2 & 3 \\ 2 & x+4 & 6 \\ 3 & 6 & x+9 \end{vmatrix}$

f(x) = $(x+1)(x^2+13x)-2(2x)+3(-3x)$

f(x) = $(x^3+13x^2+x^2+13x)-13x$

f(x) = x^3+14x^2

f'(x) = $3x^2 + 28x$

f'(x) =x(3x+28)

for decreasing f'(x) < 0

x(3x+28)< 0

The function is decreasing on $\left[-\frac{28}{3}, 0\right]$

Sol.125. (c)

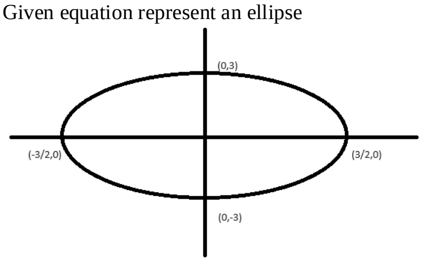
f''(x) = 6x + 28

f''(x) > 0 for x = 0

So function will attain minima at x = 0

Sol.126. (b)

Given equation represent an ellipse



From above diagram we can say that maximum value of y is 3 at x = 0

Sol.127. (a)

Let z = xy

$4x^2 + y^2 = 9$

$$y = \sqrt{9 - 4x^2}$$

$$z = xy = x\sqrt{9 - 4x^2}$$

Diff w.r.t. x

$$\frac{dz}{dx} = \sqrt{9-4x^2} - \frac{8x^2}{2\sqrt{9-4x^2}} = 0$$

$$= \sqrt{9-4x^2} - \frac{4x^2}{\sqrt{9-4x^2}} = 0$$

$$\Rightarrow \frac{9-4x^2-4x^2}{\sqrt{9-4x^2}} = 0$$

$$\Rightarrow 8x^2 - 9 = 0 \quad \Rightarrow x = \frac{3}{2\sqrt{2}}$$

Than $\Rightarrow y = \frac{3}{\sqrt{2}}$

$\Rightarrow xy = \frac{9}{4}$

Sol.128. (c)

$y = \sqrt{2-x} + \sqrt{2+x}$

$\frac{dy}{dx} = \frac{-1}{2\sqrt{2-x}} + \frac{1}{2\sqrt{2+x}} = 0$

$\Rightarrow \frac{1}{\sqrt{2+x}} = \frac{1}{\sqrt{2-x}}$

$\Rightarrow \frac{1}{2+x} = \frac{1}{2-x} \Rightarrow x = 0$

Maximum value at x = 0 that is $2\sqrt{2}$

Sol.129. (c)

its an essential condition that at maxima or minima f '(x) = 0 , and for maxima f '(x) < 0, and for minima f '(x) > 0

Sol.130. (b)

y = 2f(x) + ax – b

y' = 2f '(x) + a

for maxima at x = 0 , y' should be zero at x = 0, its only possible when a = 0 , and f '(0) is already zero given in above question.

Sol.131. (d)

for neither maxima nor minima f '(x) ≠ 0

f '(x) = $16x^2 - 8bx + 1 \neq 0$

discriminant should be negative

$64b^2 - 4(16) < 0$

$b^2 - 1 < 0$

$-1 < b < 1$

be is non negative given in question

$0 \leq b < 1$

Sol.132. (c)

$f(x) = \frac{x}{\ln x}; (x > 1)$

$f'(x) = \frac{\ln x - 1}{(\ln x)^2}$

for x > e , f '(x) > 0, so f(x) is increasing.

statement 1 is correct.

for x < e , f '(x) < 0, so f(x) is decreasing.

statement 2 is incorrect.

we know that if from interval a to b (where a < b), function is increasing than f(b) > f(a)

f(9) > f(7)

$\frac{9}{\ln 9} > \frac{7}{\ln 7}$

9ln7 > 7 ln9

statement 3 is correct

Sol.133. (d)

$f(x) = \frac{x}{\ln x}; (x > 1)$

$f'(x) = \frac{\ln x - 1}{(\ln x)^2} = 0$

than lnx – 1 = 0

lnx = 1

x = e

$f'(x) = \frac{\ln x - 1}{(\ln x)^2} = \frac{1}{\ln x} - \frac{1}{(\ln x)^2}$

$f''(x) == -\frac{1}{x(\ln x)^2} + \frac{2}{x(\ln x)^3}$

$f''(e) == -\frac{1}{e} + \frac{2}{e} = \frac{1}{e} > 0$

so minima at x = e.

and minimum value will be f(e) = e

Sol.134. (b)

f(x) = cos2x + x

f '(x) = – 2 sin2x + 1 = 0

sin2x = ½

2x = π/6 or 5π/6

x = π/12 or 5π/12

f "(x) = – 4cos2x

f "(π/12) = – 4cos (π/6) = negative

so maxima at x = π/12

maximum value

f(π/12) = $\frac{\sqrt{3}}{2} + \frac{\pi}{12}$

Sol.135. (a)

f(x) = cos2x + x

f '(x) = – 2 sin2x + 1 = 0

sin2x = ½

2x = π/6 or 5π/6

x = π/12 or 5π/12

f "(x) = – 4cos2x

f "(5π/12) = – 4cos (5π/6) = positive

so local minima at x = 5π/12

local minimum value

f(π/12) = $\frac{\sqrt{3}}{2} + \frac{\pi}{12}$

at extreme values

f(–π/2) = $-1 + \frac{\pi}{2}$, f(π/2) = $-1 - \frac{\pi}{2}$

so absolute minimum value is $-1 - \frac{\pi}{2}$

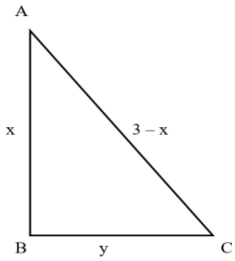
Sol.136. (b)

$x^2 + 2x + 11 = (x + 1)^2 + 10$

minimum value of $(x + 1)^2 + 10$ is 10

so minimum value of $\log[(x + 1)^2 + 10]$ is log10 or 1

Sol.137. (c)



By Pythagoras theorem

$x^2 + y^2 = (3 - x)^2$

$x^2 + y^2 = 9 - 6x + x^2$

$x = \frac{9 - y^2}{6}$ ……(i)

Area of right angled triangle

$A = \frac{1}{2}xy = \frac{9 - y^2}{12}y = \frac{9y - y^3}{12}$

For maximum area

$\frac{dA}{dy} = \frac{9 - 3y^2}{12} = 0$

$y = \sqrt{3}$

Now by equation (i)

x = 1

By triangle ABC

$\tan A = \frac{\sqrt{3}}{1} \Rightarrow A = \frac{\pi}{3}$

Sol.138. (a)

By above solution $A = \frac{1}{2}(1)(\sqrt{3}) = \frac{\sqrt{3}}{2}$

Sol.139. (d)

$f(x) = x^2 + 9$

f '(x) = 2x

for x > 0 function is increasing and for x < 0 function is decreasing

f''(x) = 2 > 0 , so function has minima at x = 0

so both statements are incorrect.

Sol.140. (c)

for x > 0

$f(x) = x^2$

f '(x) = 2x > 0 for all x > 0

for x < 0

$f(x) = -x^2$

f '(x) = – 2x > 0 for all x < 0

so f '(x) is always positive so function is always increasing.

At x = 0, right hand and left hand both derivatives are zero, so function is differentiable.

Sol.141. (b)

$f(x) = 1 - \sqrt[3]{(x-1)^2}$

f(1) = 1

Right hand limit at x = 1

$f(1+0) = \lim_{h \to 0} f(1+h) = 1 - (1+h-1)^{\frac{2}{3}} = \lim_{h \to 0}\left(1 - h^{\frac{2}{3}}\right)$

i.e. less than 1

Left hand limit at x = 1

$f(1-0) = \lim_{h \to 0} f(1-h) = 1 - (1-h-1)^{\frac{2}{3}} = \lim_{h \to 0}\left(1 - h^{\frac{2}{3}}\right)$

i.e. less than 1

by above limits we can say that there is a maxima at x = 1.

Sol.142. (a)

A wire of length 20 cm is to be bent into a rectangle.

2(*l* + b) = 20

l + b = 10

we know that rectangle has largest area is square.

[proof given in ncert question]

for square *l* = b = 5

so maximum area = 25 sq. units

Statement 1 is correct and 2 is wrong.

Sol.143. (c)

$f(x) = \frac{1}{x-5}$

$f'(x) = -\frac{1}{(x-5)^2} < 0$

f'(x) < 0 so function is always decreasing in its domain.

Statement I is correct and II is wrong.

Sol.144. (c)

$f(x) = \frac{x^3 + 128}{x} = x^2 + \frac{128}{x}$

$f'(x) = 2x - \frac{128}{x^2} = 0$

x = 4

$f''(x) = 2 + \frac{256}{x^3} \Rightarrow f''(4) > 0$

So function has minima at x = 4

And minimum value y = 48

Minima at x = 4 means slope of curve changing from negative to positive, so statement 2 is wrong.